

EEE 4483

Digital Electronics and Pulse Techniques

Lecture 1: Review of Electronic Devices

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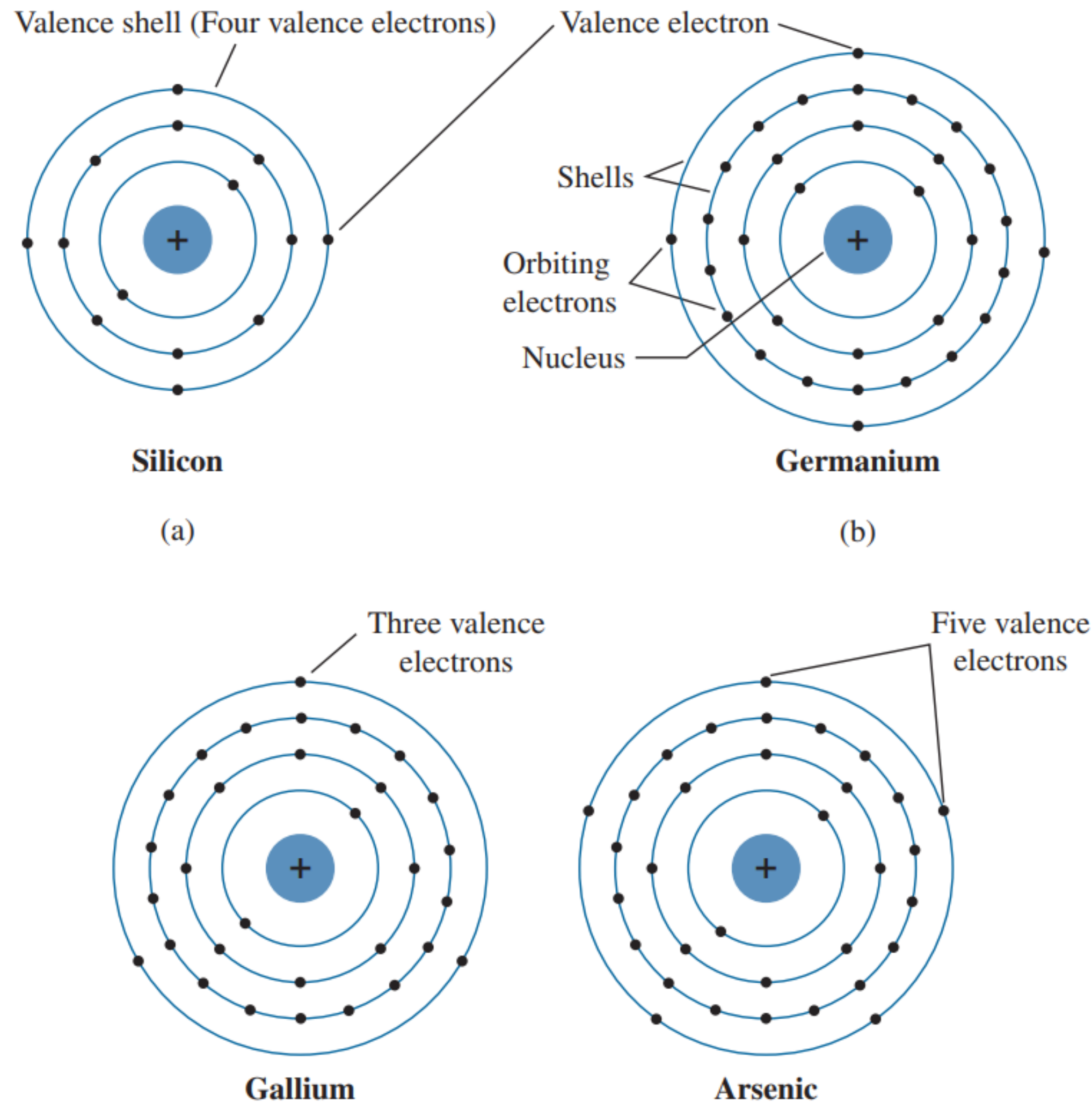
Referred Textbook:

Electronic Devices and Circuit Theory, 11th edition
by Robert L. Boylestad and Louis Nashelsky

Microelectronic Circuits, 6th edition
by Adel S. Sedra and Kenneth C. Smith

Microelectronics: Circuit Analysis and Design, 4th edition
by Donald A. Neamen





- **Semiconductors** are a special class of elements having a conductivity between that of a good conductor and that of an insulator
- At room temperature or when heated, some **valence electrons** gain enough energy to jump across the bandgap into the **conduction band**
- These free electrons in the conduction band, and the holes left behind in the valence band, enable current flow — acting like a conductor
 - > **Intrinsic Semiconductor:** Highly pure form (very low impurity levels)
 - > **Intrinsic Carriers:** Electrons/holes generated only by thermal energy

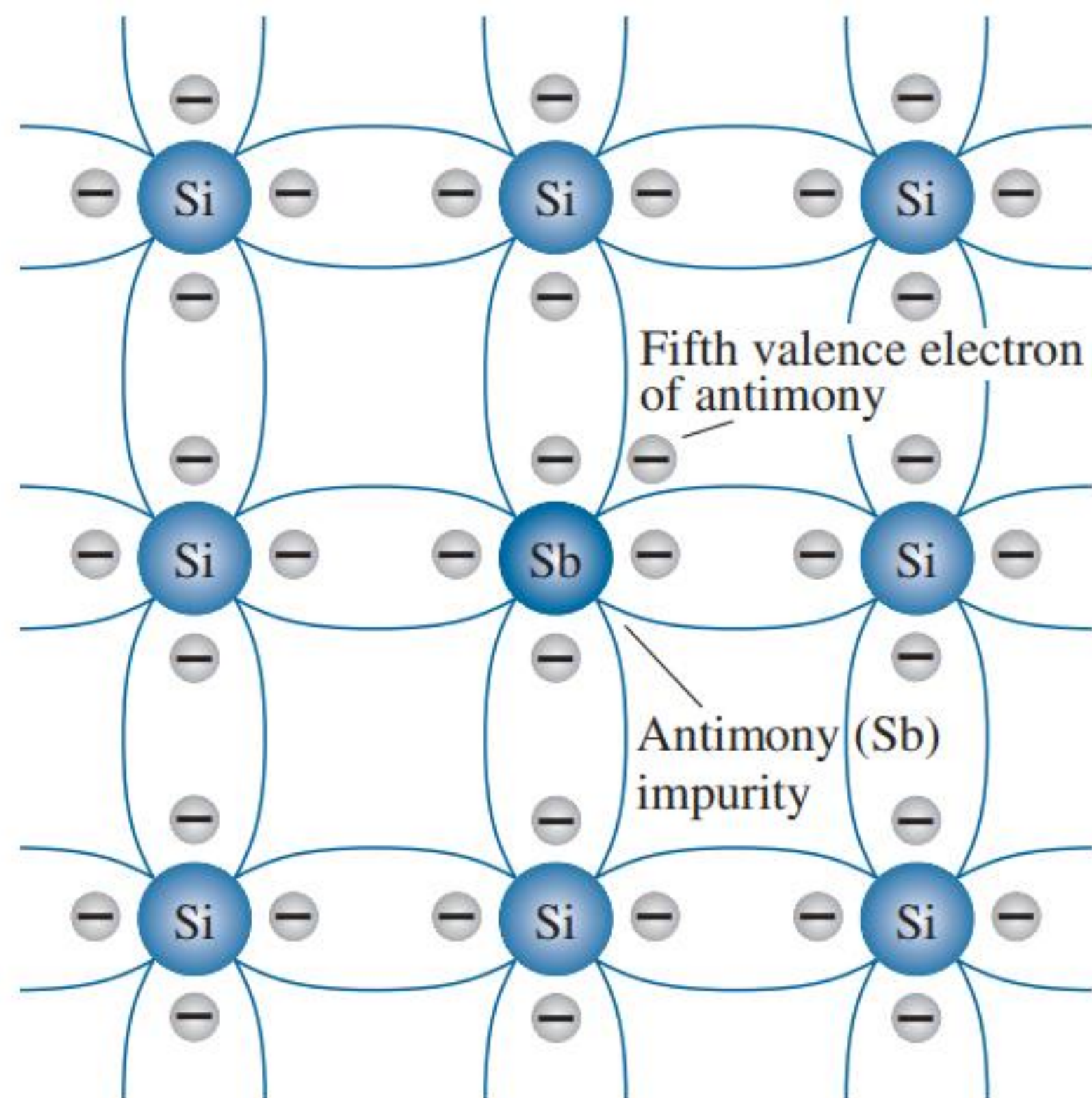


Fig: Antimony impurity in n-type material

n-Type Materials

- Formed by doping pure semiconductor with **Group V** elements (e.g., P, As, Sb) that have five valence electrons
- Each dopant forms four covalent bonds, leaving one free electron that is loosely bound and easily moves
- These pentavalent atoms are called **donor atoms** because they donate free electrons to the conduction band
- The free electrons are the majority carriers, enabling high conductivity in n-type material

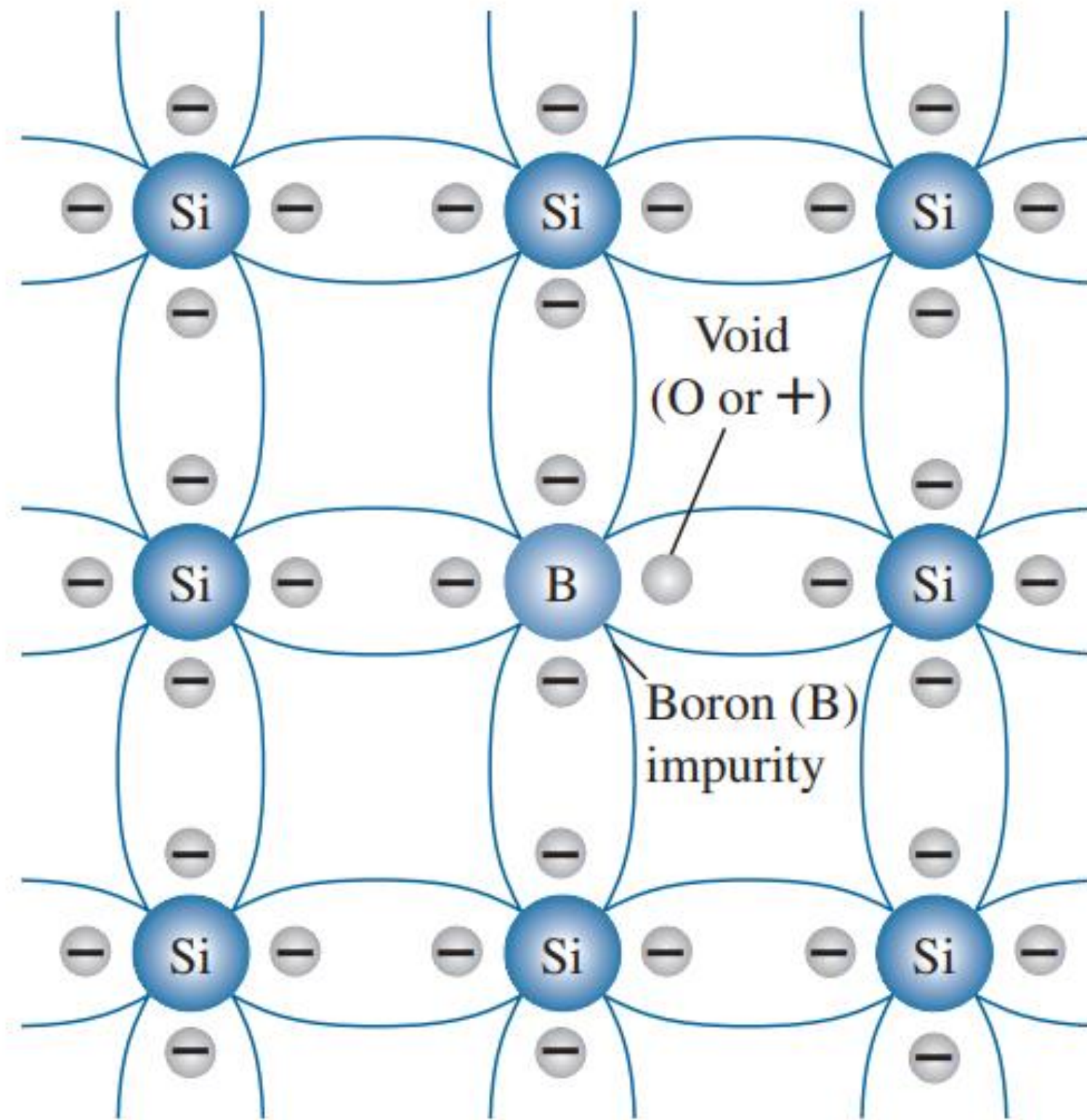


Fig: Boron impurity in p-type material

p-Type Materials

- Formed by doping pure semiconductor with **Group III** elements (e.g., B, Al, Ga) that have three valence electrons
- Each dopant forms three covalent bonds, leaving one bond incomplete, creating a **hole**
- These trivalent atoms are called **acceptor atoms** because they accept electrons to fill the missing bond
- The holes are the majority carriers, enabling high conductivity in p-type material

- In n-type material –

- › Doping adds many **electrons** (via donor atoms), making electrons the **majority carriers**.
- › Thermal generation still creates electron-hole pairs → **holes** are present but in very small numbers → **minority carriers**.
- › **Majority carriers (electrons)** = Electrons from donor atoms (doping) + electrons from thermal generation

- In p-type material –

- › Doping adds many **holes** (via acceptor atoms), making holes the **majority carriers**.
- › Thermal generation still creates electron-hole pairs → **electrons** are present in small numbers → **minority carriers**.
- › **Majority carriers (holes)** = Holes from acceptor atoms (doping) + Holes from thermal generation

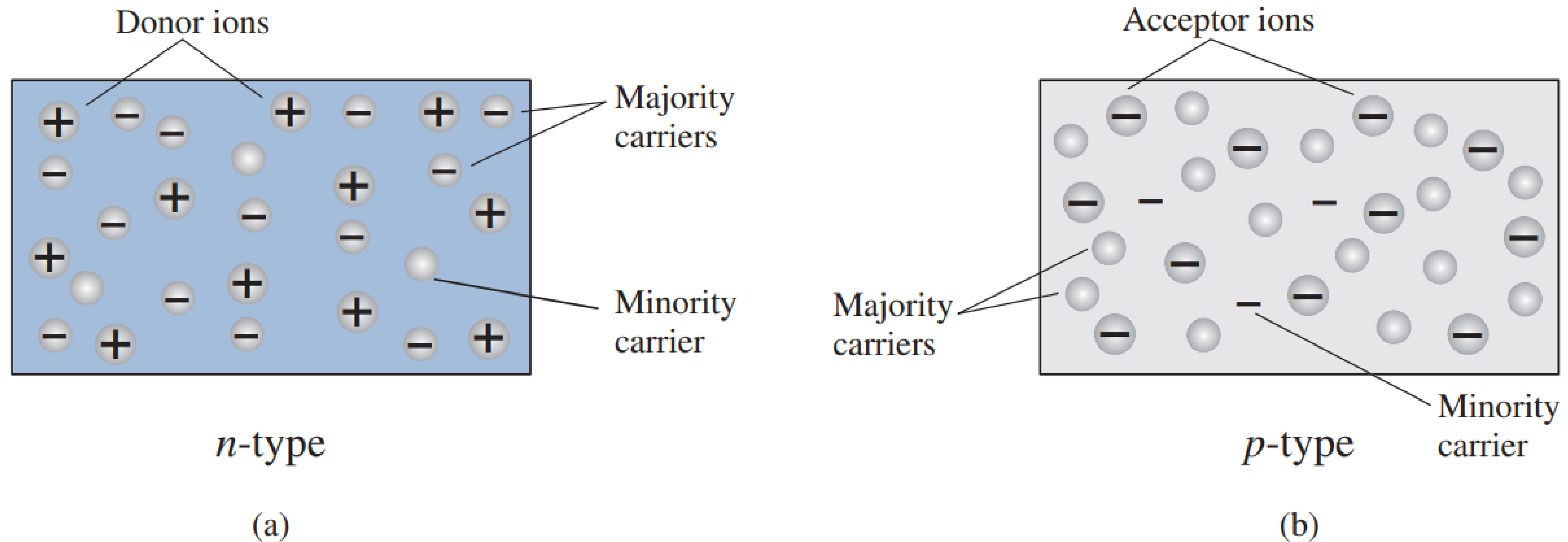


Fig: (a) *n*-type material, (b) *p*-type material

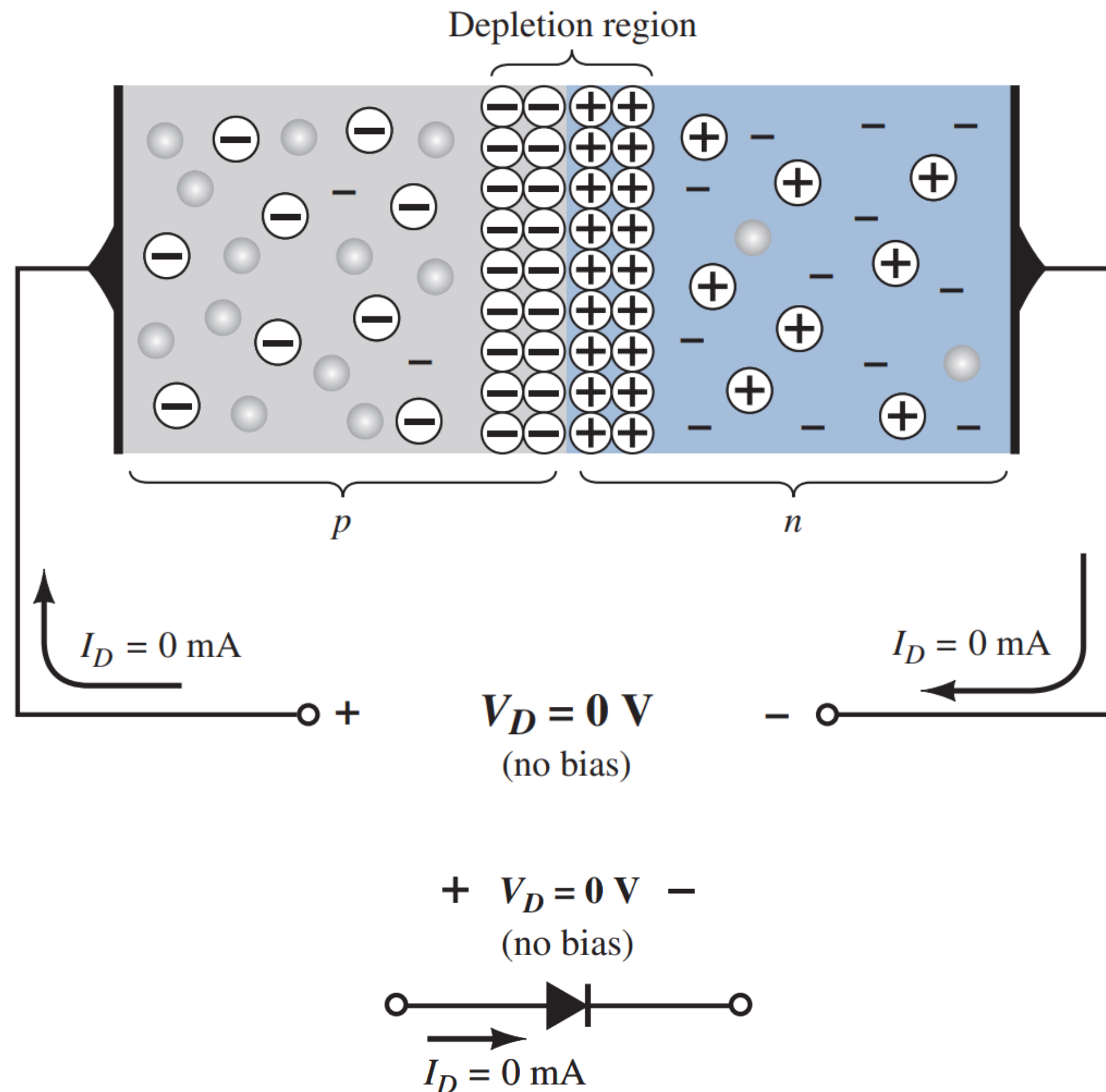


Fig: A p–n junction with no external bias

No Applied Bias (Voltage = 0 V)

- When **p-type** and **n-type** materials join, **electrons** diffuse from **n** to **p**, and **holes** from **p** to **n**, due to carrier **concentration** differences
- This diffusion causes recombination near the junction, depleting free carriers
- Left behind: **immobile ions** (negative on p-side, positive on n-side)
- An **electric field** builds up → opposes further diffusion
- System reaches equilibrium when –
diffusion current = drift current

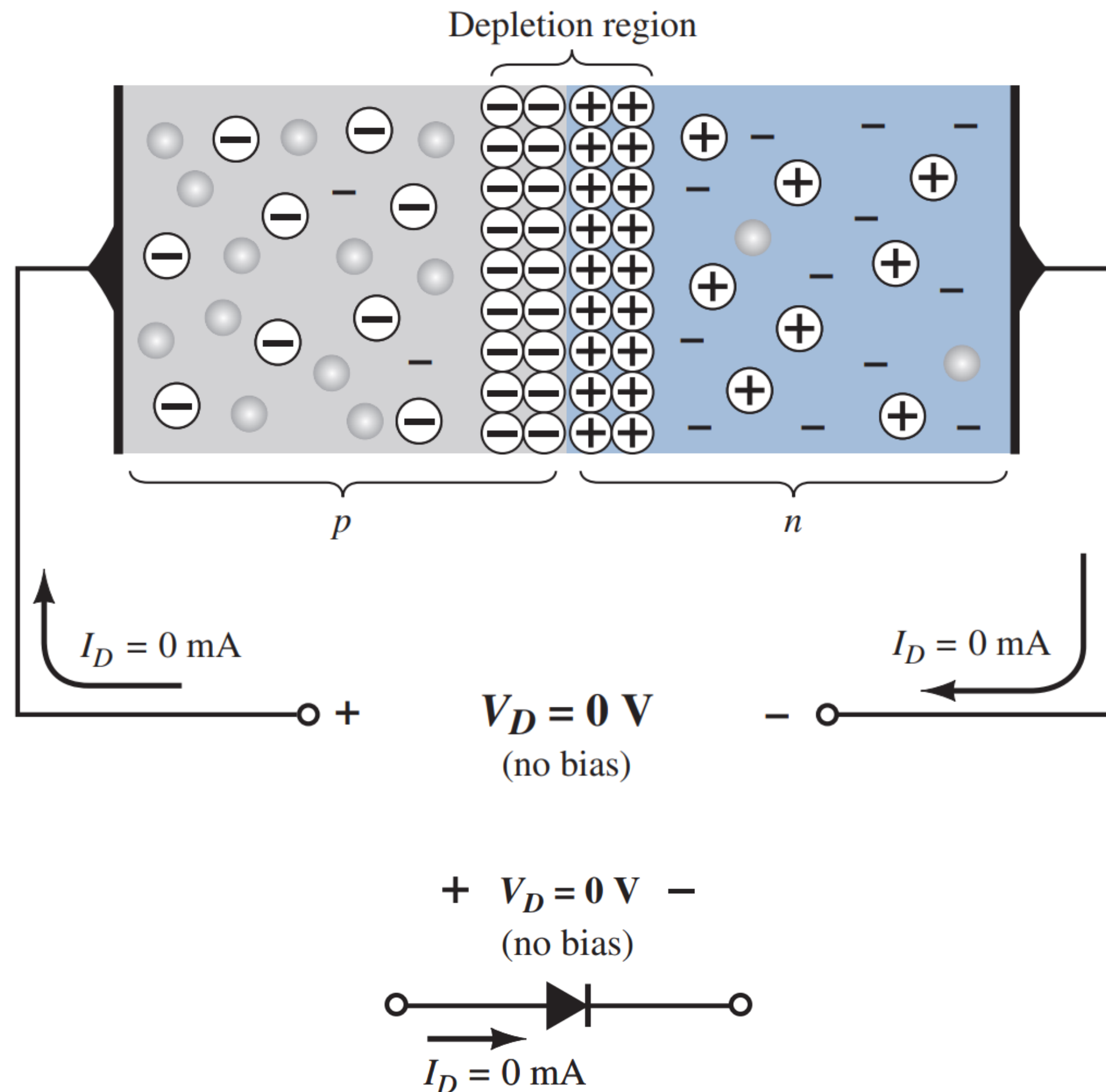


Fig: A p–n junction with no external bias

■ Diffusion Current

- › Caused by **concentration gradient** of **majority carriers**
- › Electrons move from high (n-side) to low (p-side) concentration
- › Holes move from p-side → n-side
- › This movement creates current from carrier movement

■ Drift Current

- › Caused by **electric field** in the depletion region
- › The field pushes electrons toward n-side, and holes toward p-side (acts on **minority carriers**)
- › It opposes the direction of diffusion

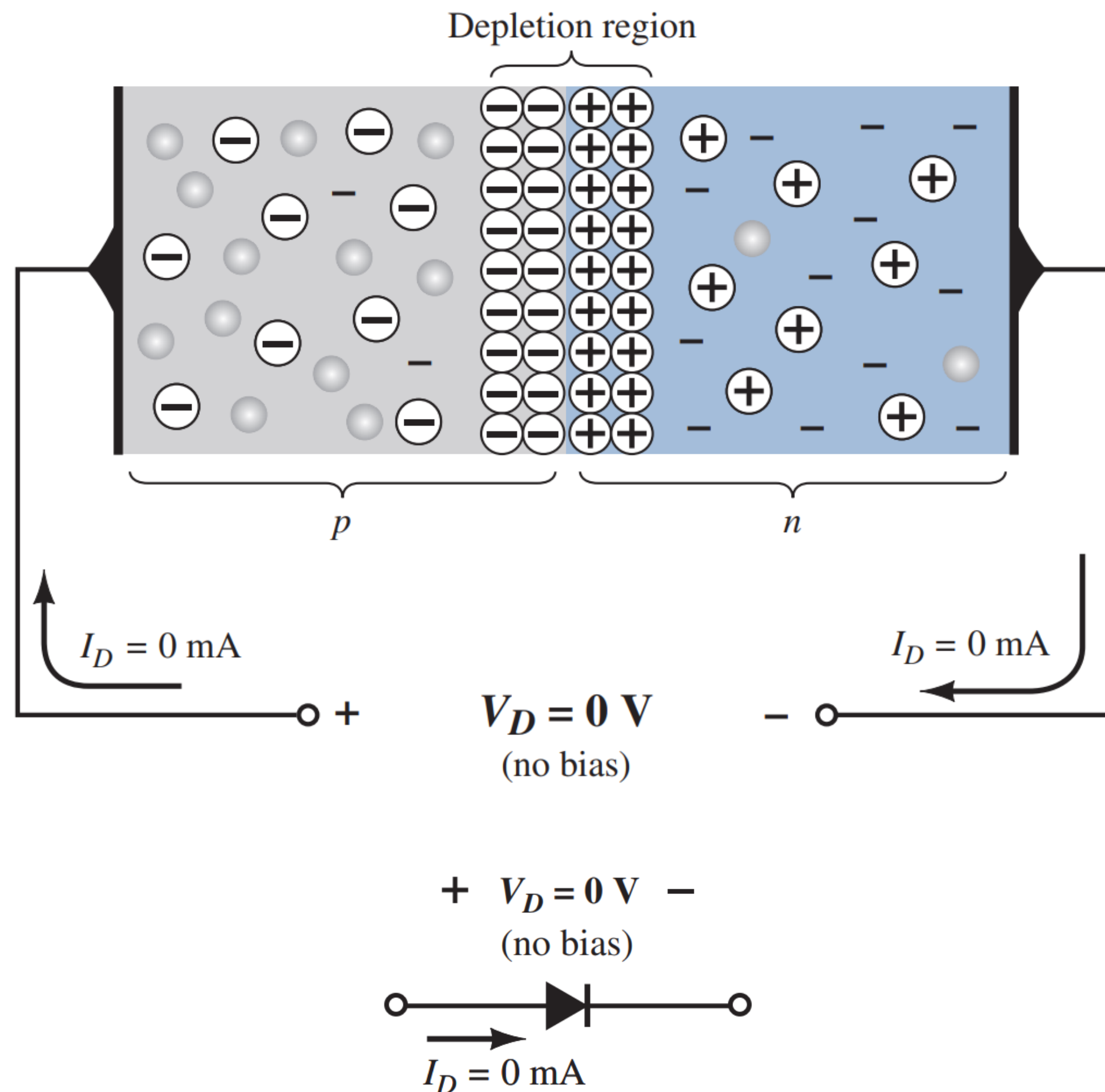


Fig: A p-n junction with no external bias

■ At Equilibrium

- › As more carriers diffuse, the electric field gets stronger
- › Eventually, this field is strong enough to pull back carriers
- › Eventually, the **drift current** (caused by this electric field acting on minority carriers) exactly balances the **diffusion current** (from majority carriers)
- › Although electrons and holes are constantly moving, the opposing flows cancel each other
- › Result: **No net current**

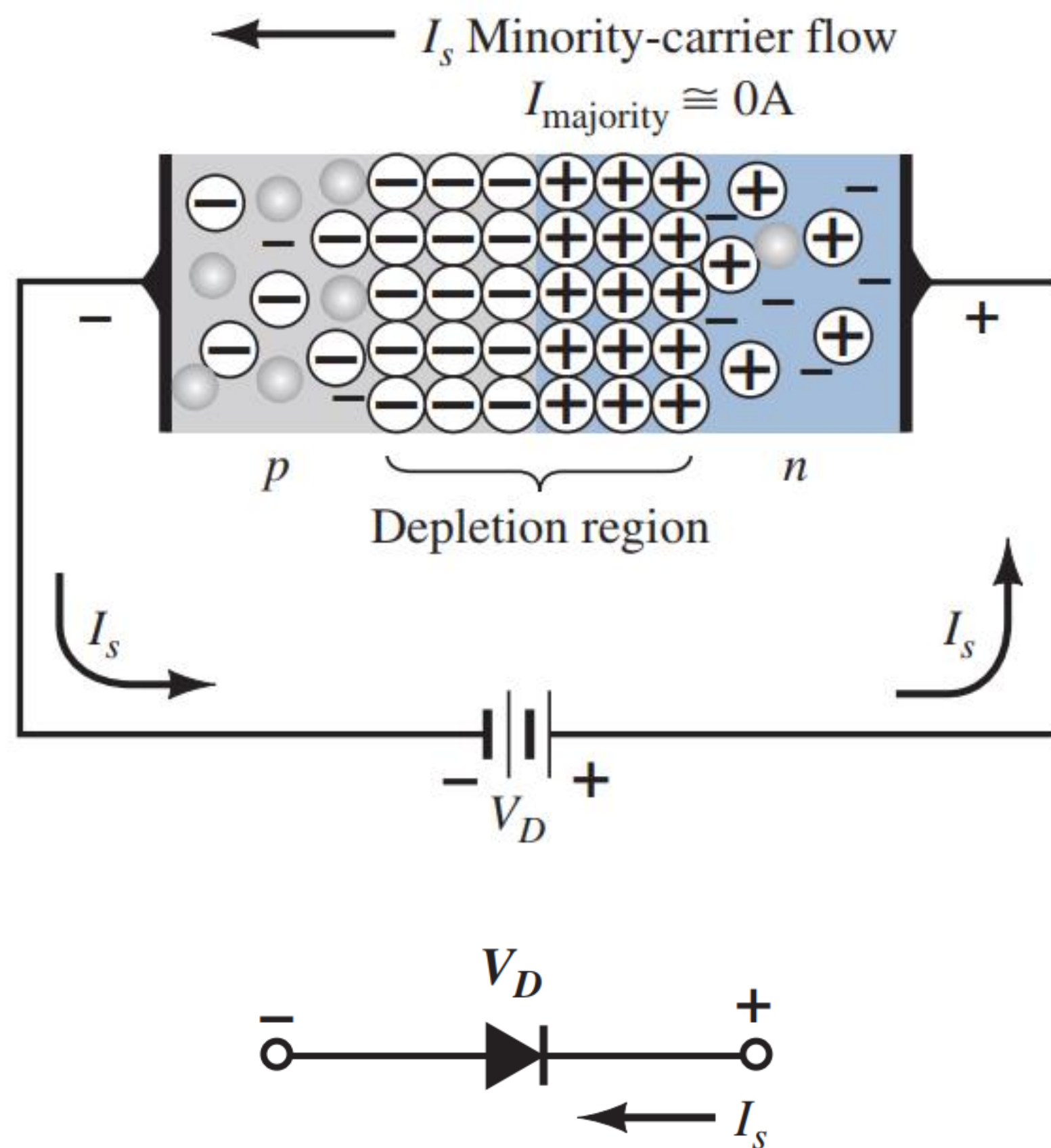


Fig: Reverse-biased p–n junction

Reverse-Bias Condition ($V_D < 0\text{ V}$)

- Positive terminal connected to n-type, negative to p-type
 - Electrons pulled toward the positive terminal
 - Holes pulled toward the negative terminal
- Depletion region **widens**
- **Majority Carrier Flow Stops**
 - > Barrier becomes too wide
 - > Majority carriers cannot cross → **flow ≈ 0**
 - **Minority Carrier Flow Continues**
 - > Minority carriers still drift across the junction
 - > Results in a small reverse current, called **reverse saturation current (I_s)**

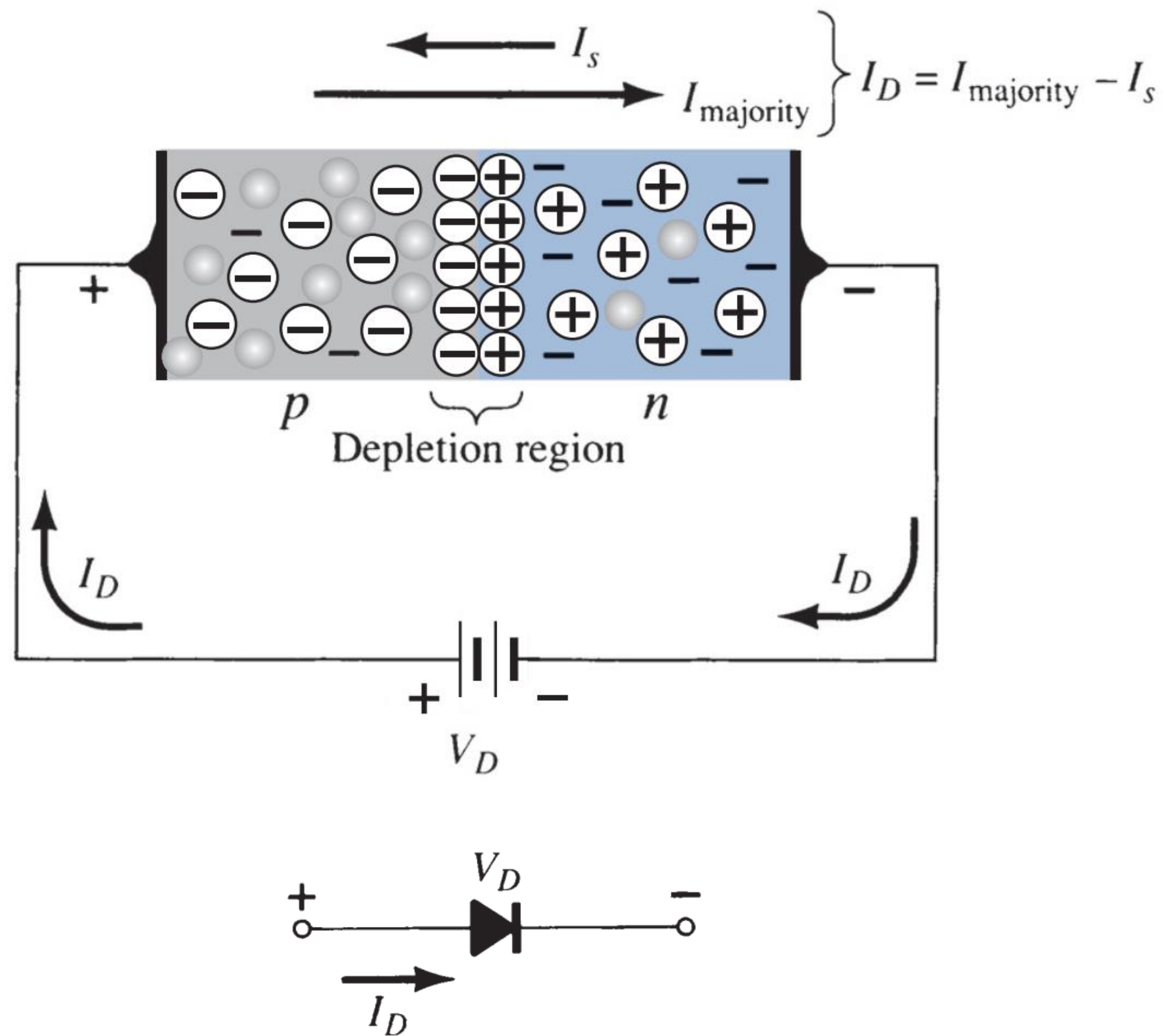


Fig: Forward-biased p–n junction

Forward-Bias Condition ($V_D > 0$ V)

- Positive terminal connected to p-type, negative to n-type
- This pushes electrons from n-side and holes from p-side toward the junction
- Depletion Region Narrows
 - › Electrons and holes recombine with ions at the junction
 - › Barrier height **reduces**, making it easier for majority carriers to cross

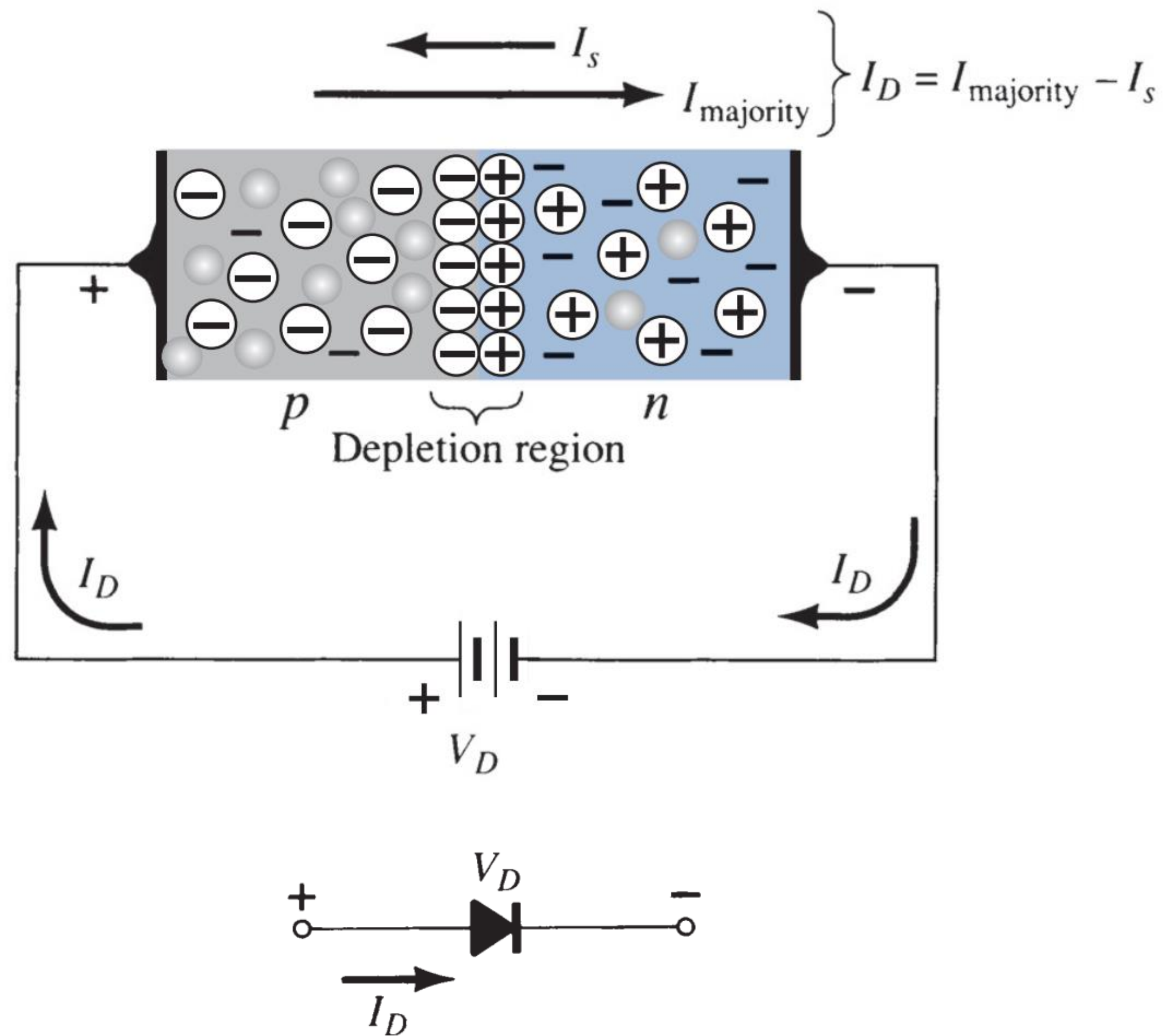


Fig: Forward-biased p–n junction

■ Majority Carrier Flow Increases

- Majority carriers now easily cross the junction
- Large current flow begins
- Minority carrier flow remains constant

■ Exponential Current Increase

- As forward voltage increases, depletion width shrinks more
- Results in exponential rise in current after threshold (knee voltage/cut-in voltage)

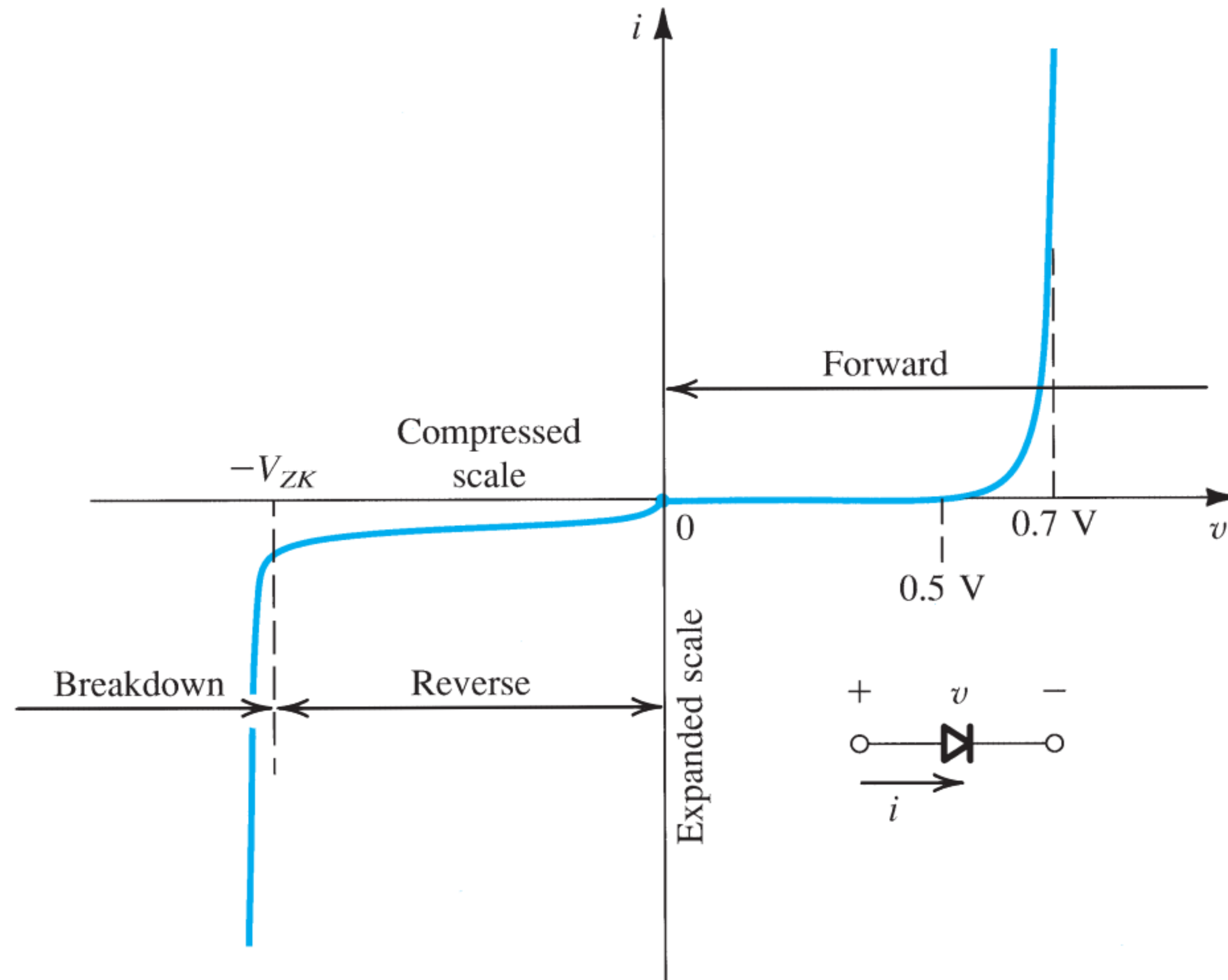


Fig: I-V characteristics of diode

Shockley's Diode Equation

The current through a diode is given by the Shockley equation –

$$I_D = I_S(e^{V_D/nV_T} - 1)$$

Where,

I_D = Diode current

I_S = Reverse saturation current

V_D = Applied forward-bias voltage

n = Ideality factor (typically 1 – 2)

V_T = Thermal voltage = $\frac{kT}{q}$

k = Boltzmann's constant = $1.38 \times 10^{-23} \text{ J/K}$

T = Absolute temperature in kelvin = $273 + ^\circ\text{C}$

q = Charge of an electron = $1.6 \times 10^{-19} \text{ C}$

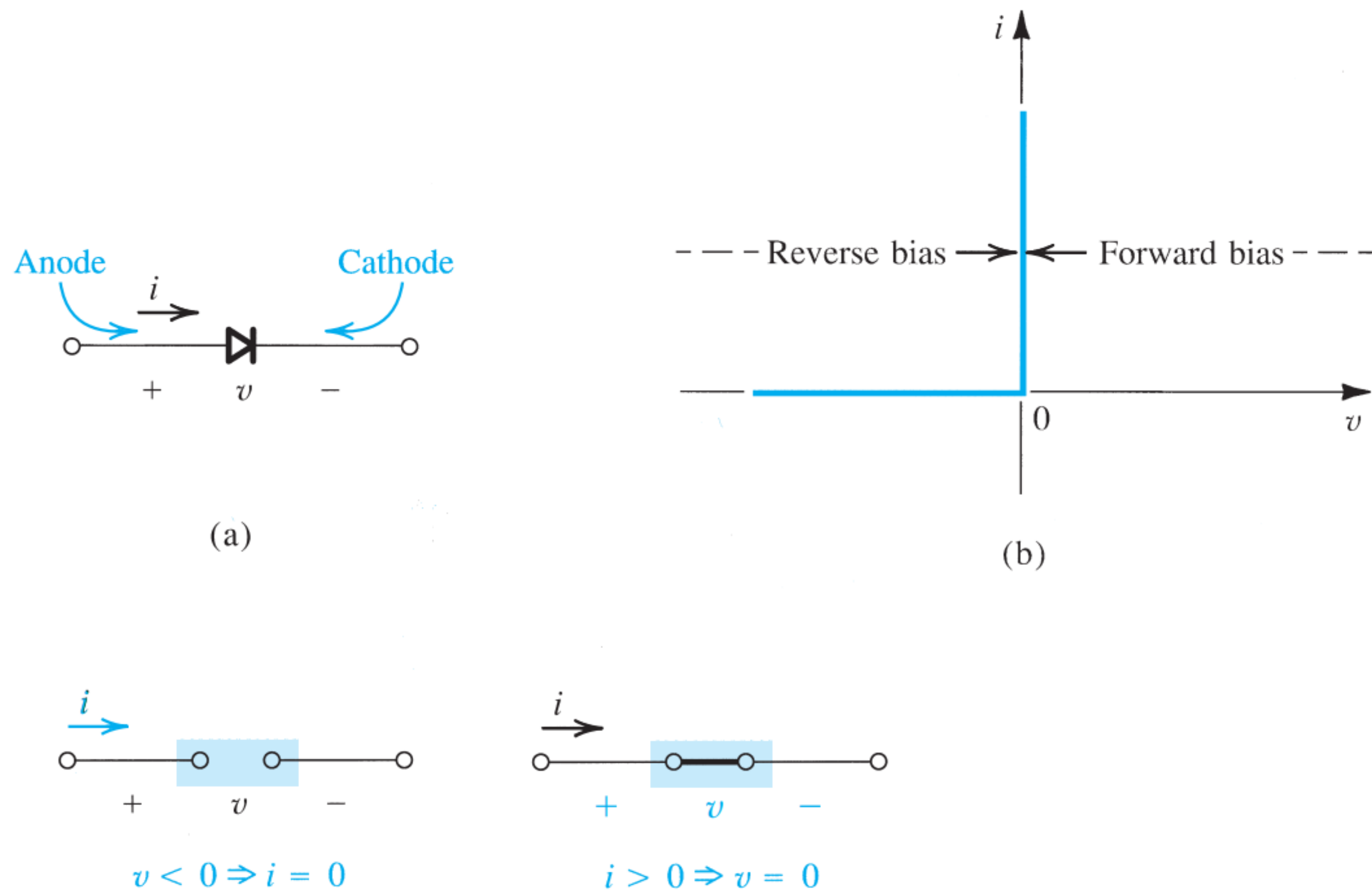


Fig: Ideal diode model

The Ideal Model

- Forward Bias (ON State)
 - > Diode acts as a perfect conductor
 - > Voltage drop across diode: $V_D = 0 \text{ V}$
 - > Current flows freely: $I_D > 0$
- Reverse Bias (OFF State)
 - > Diode acts as a perfect insulator
 - > No current flow: $I_D = 0$, regardless of reverse voltage

This model is best for basic switching behavior and introductory analysis

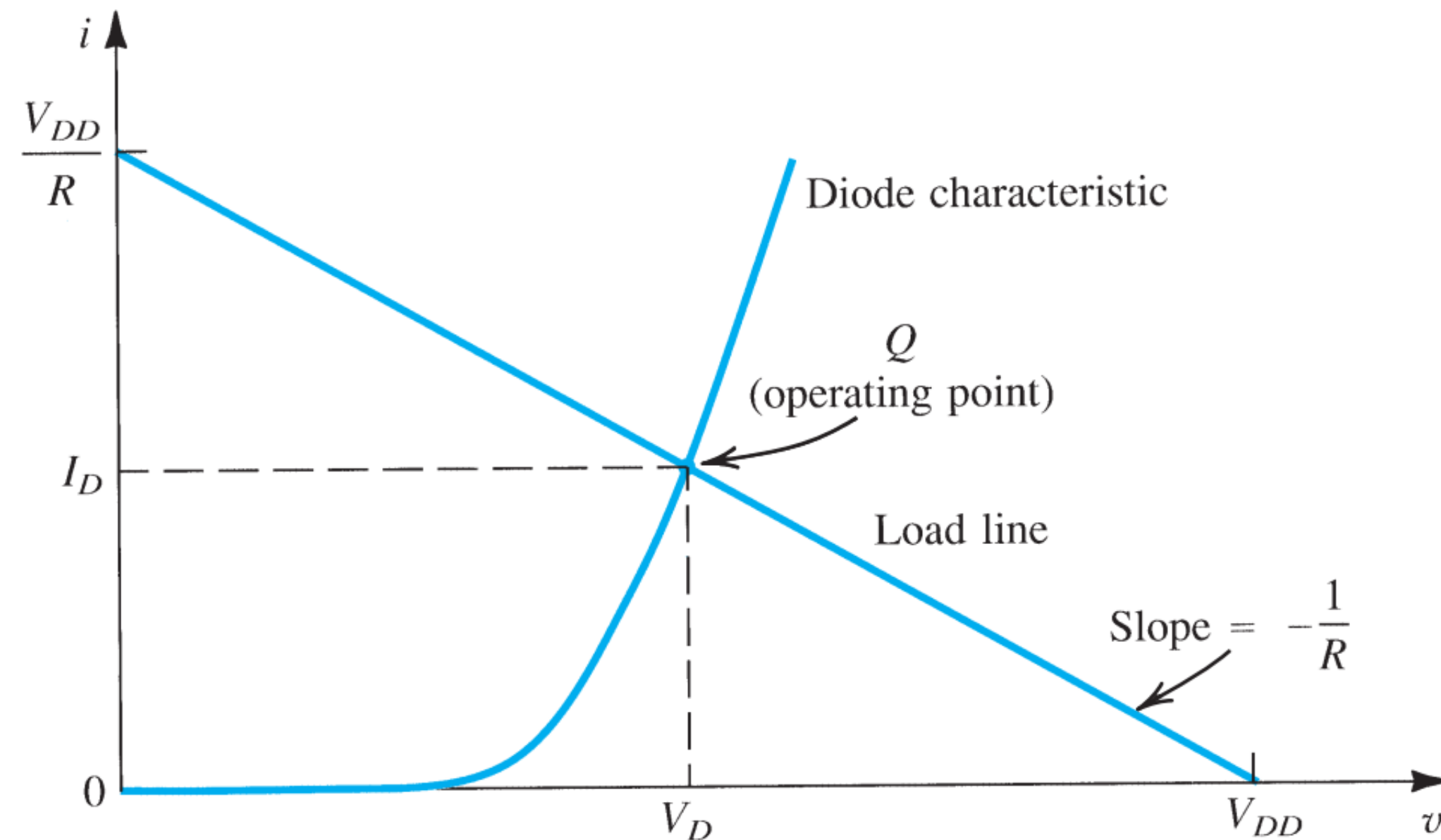
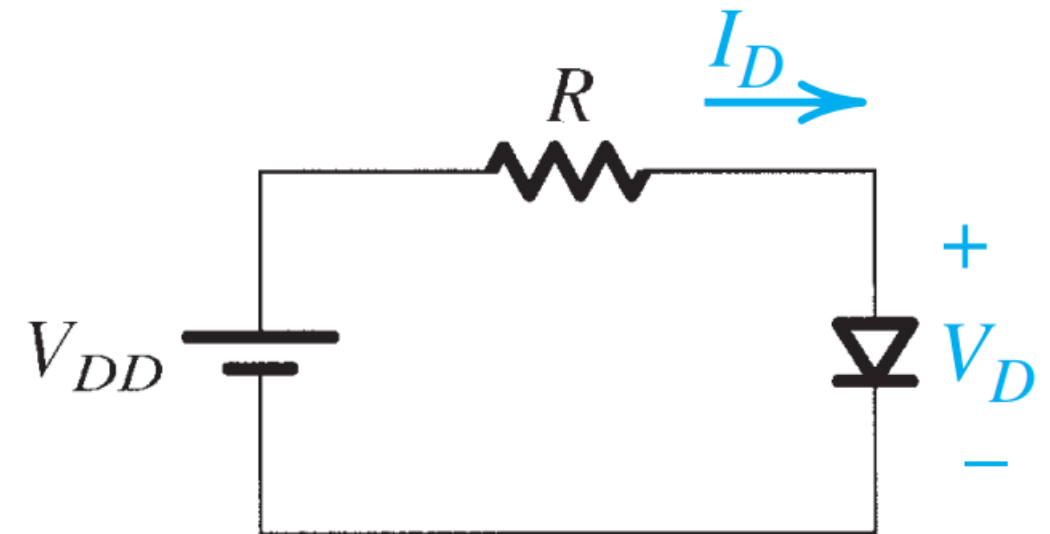


Fig: Exponential model of diode

The Exponential Model

Shockley's Diode Eqn. (approximated) –

$$I_D = I_S e^{V_D/V_T}$$

- > Diode model is generated by following this equation
- > Most accurate model for diode behavior
- > Highly nonlinear, making it difficult to solve analytically

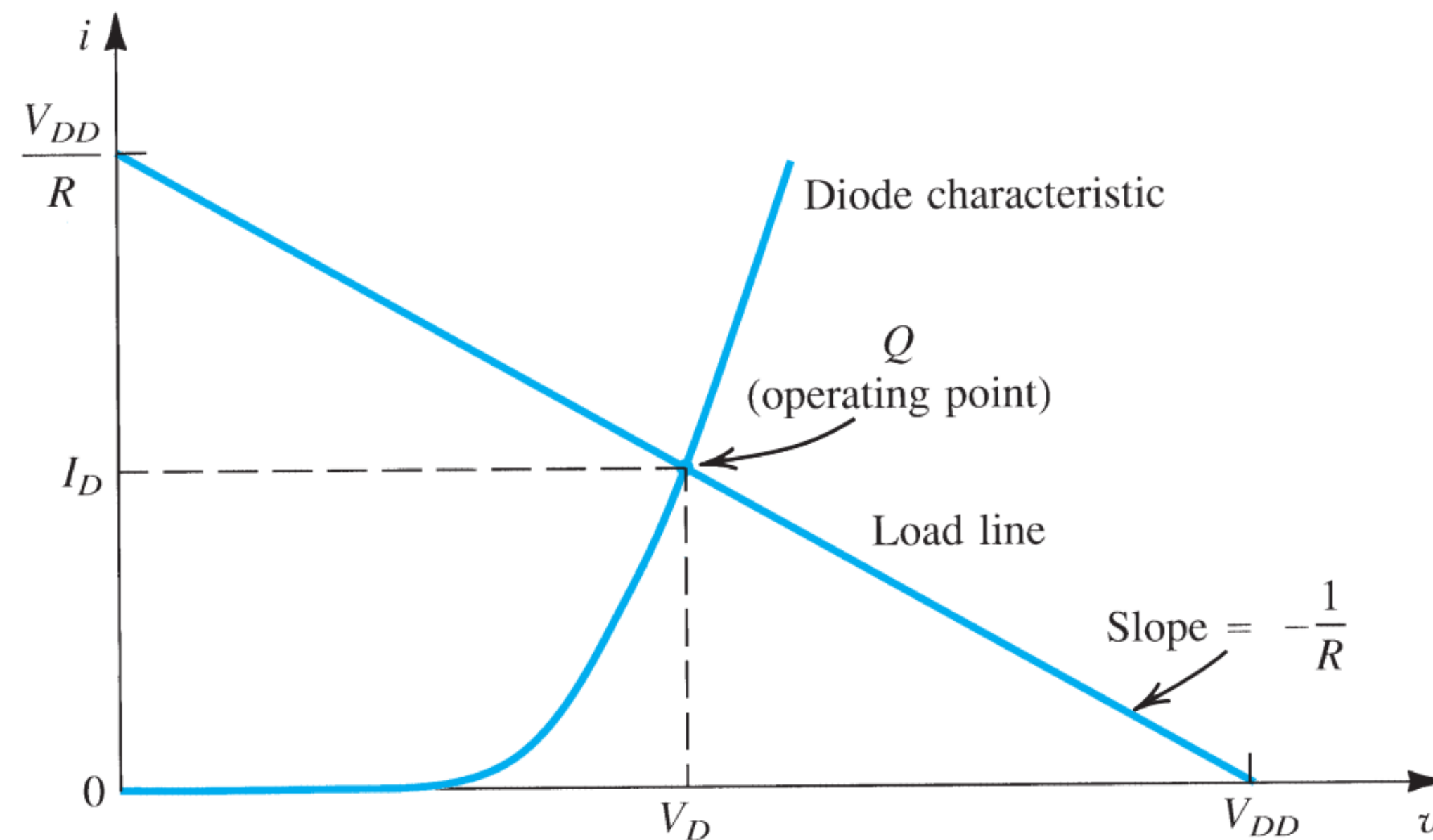
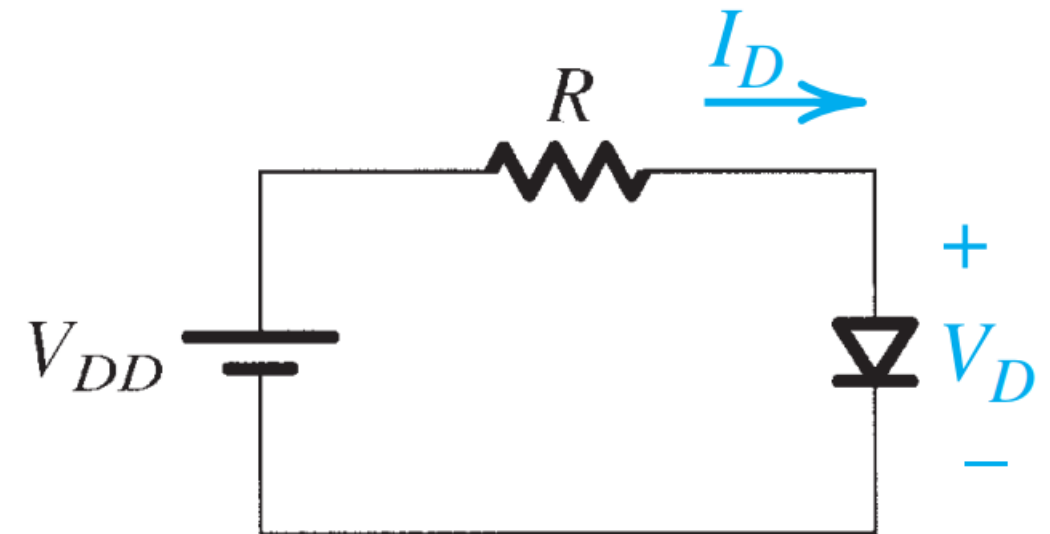


Fig: Exponential model of diode

The Exponential Model

Step 1: Circuit Analysis

- › Using KVL in the circuit at the top –

$$I_D = \frac{V_{DD} - V_D}{R}$$

- › Now we have two equations –

Diode equation: $I_D = I_S e^{V_D/V_T}$

Load line equation: $I_D = \frac{V_{DD} - V_D}{R}$

- › Two unknowns: I_D and V_D

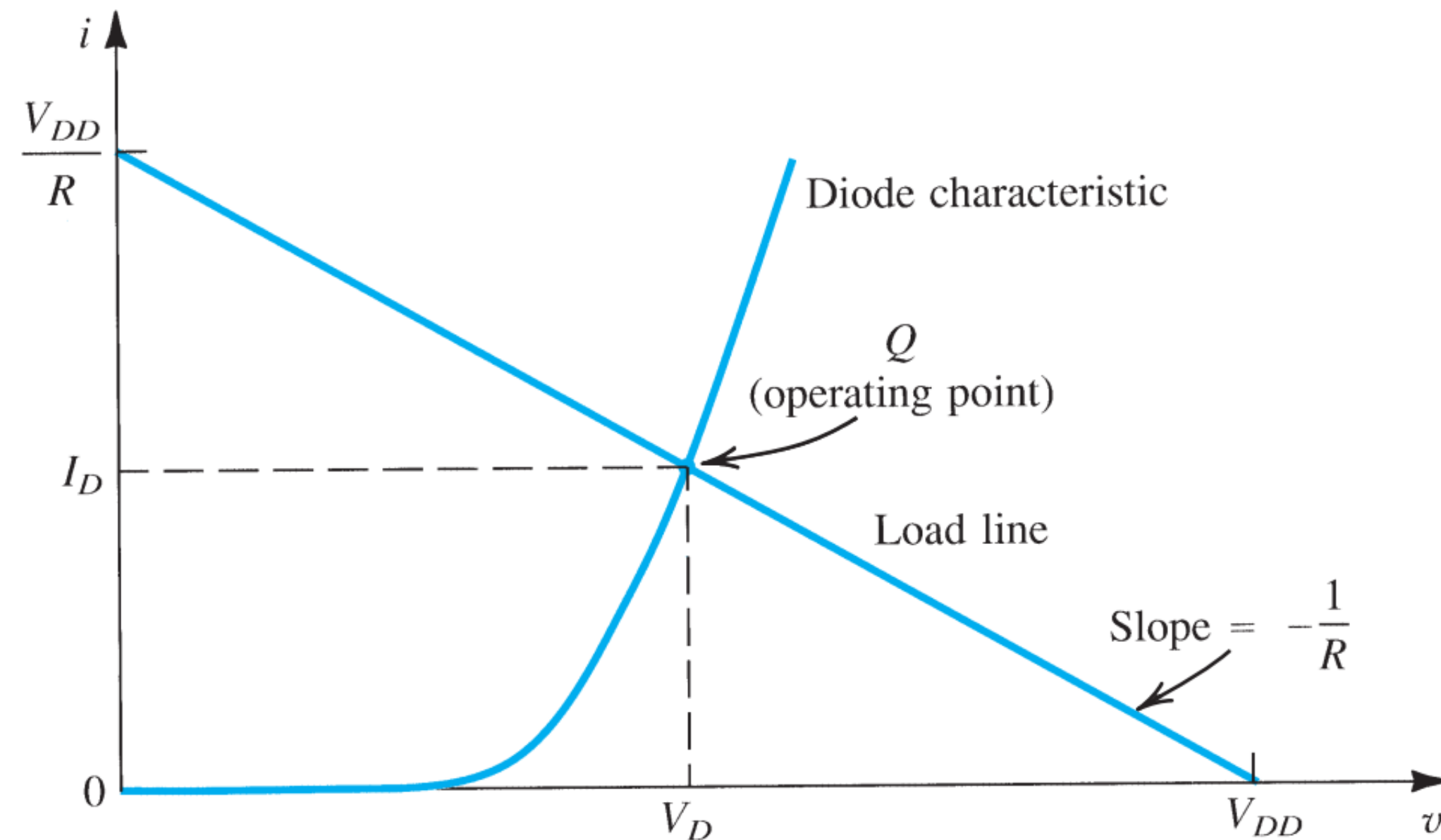
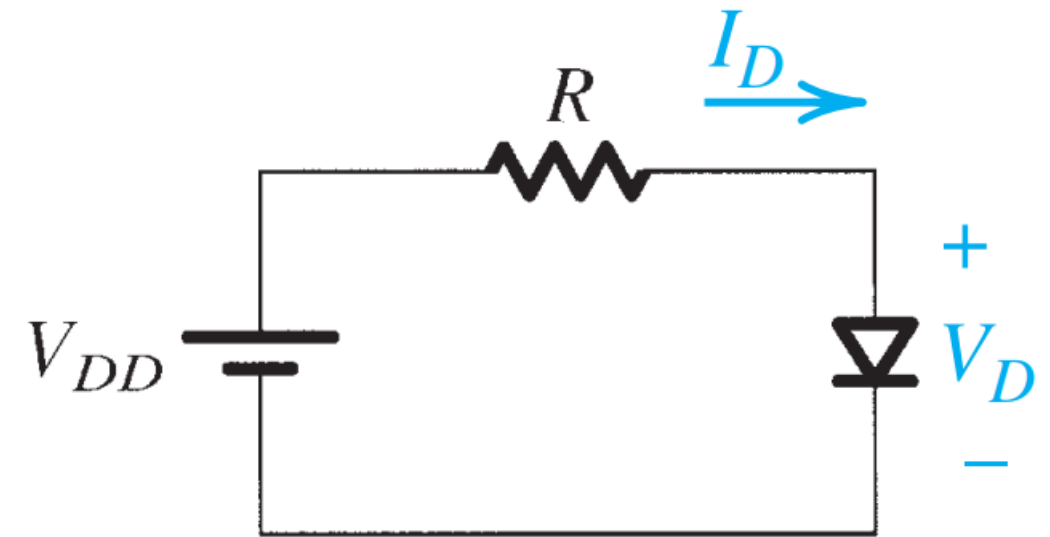


Fig: Exponential model of diode

The Exponential Model

Step 2: Graphical Analysis

- › Plot both equations on the I-V plane –
Exponential Curve → Diode equation
Straight Line → Load line
- › Operating Point (Q-point) is the intersection of the two curves
- › Q-point gives actual values of I_D and V_D

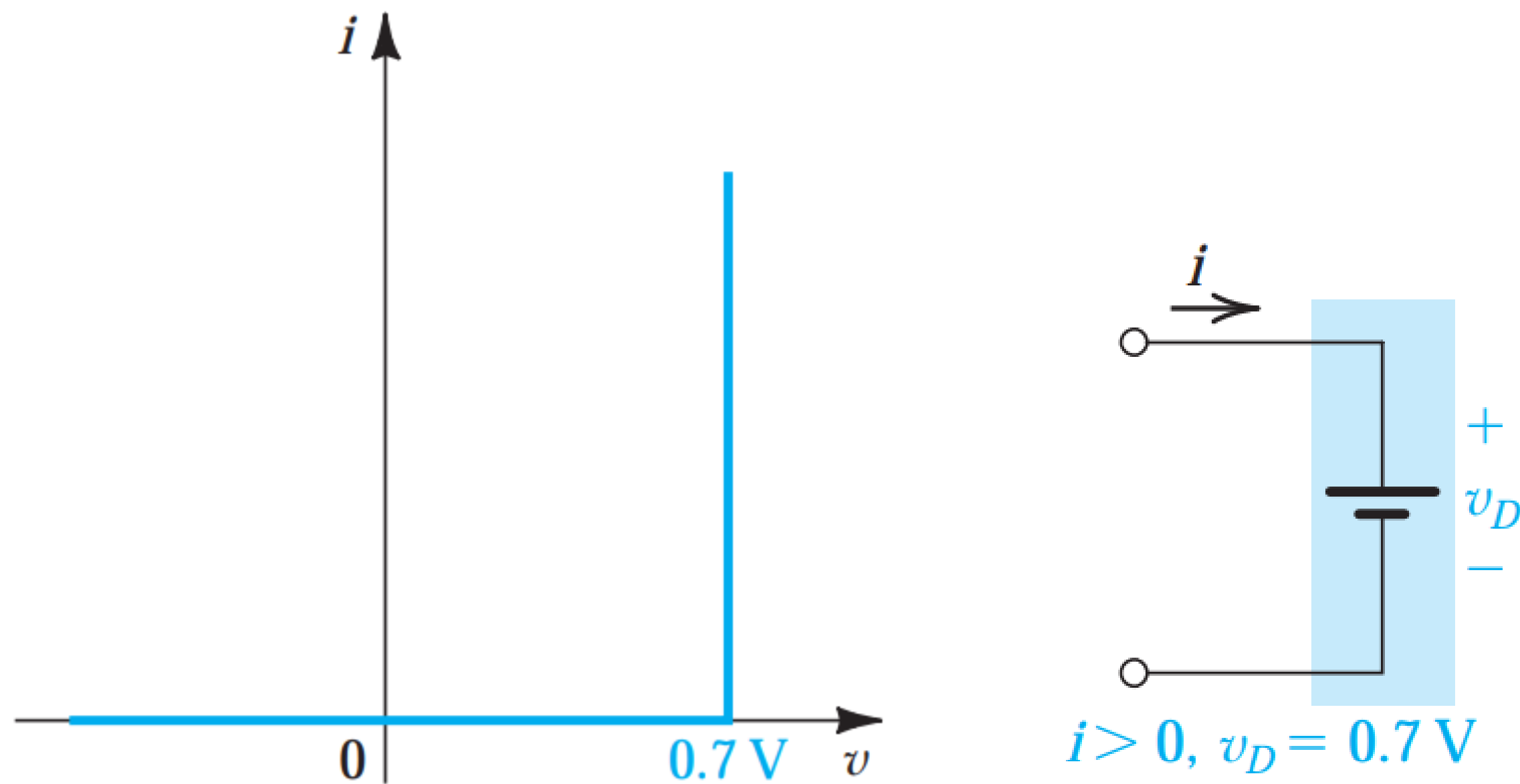


Fig: Constant-Voltage-Drop model of diode

The Constant-Voltage-Drop Model

- Assumes a fixed voltage drop across a forward-biased diode –

$$V_D \approx 0.7\text{ V (for Silicon)}$$

- Based on practical observation –
 - Actual forward voltage ranges from 0.6 V to 0.8 V
 - Simplified to a constant value for ease of analysis

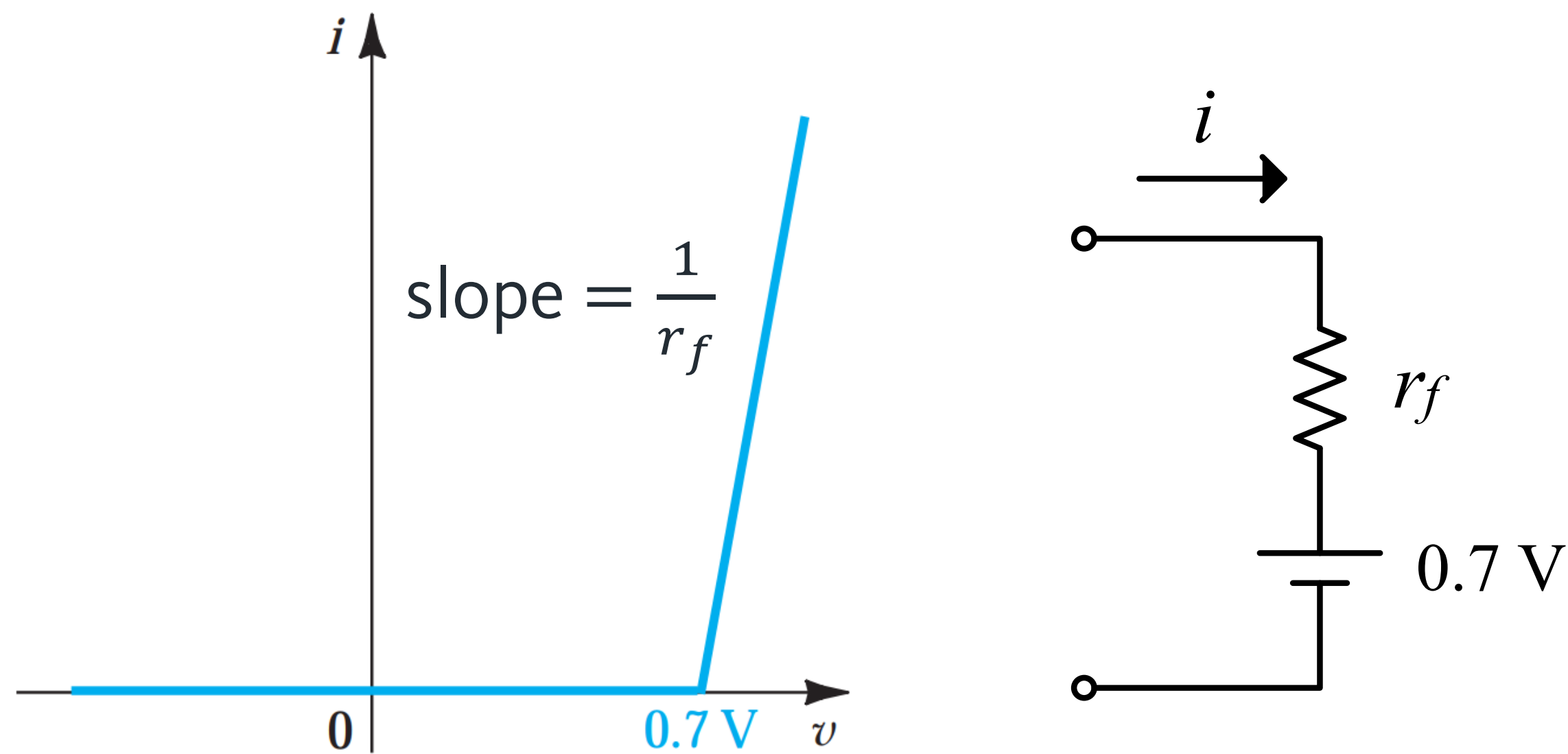


Fig: Piecewise Linear model of diode

Piecewise Linear Model

Improves upon the constant-voltage-drop model by adding a small forward resistance.

Diode is modeled as –

$$V_D = 0.7 + I_D \cdot r_f \text{ (for silicon)}$$

Here, r_f is small resistance when the diode is ON (typically a few ohms)

Bipolar Junction Transistor (BJT)

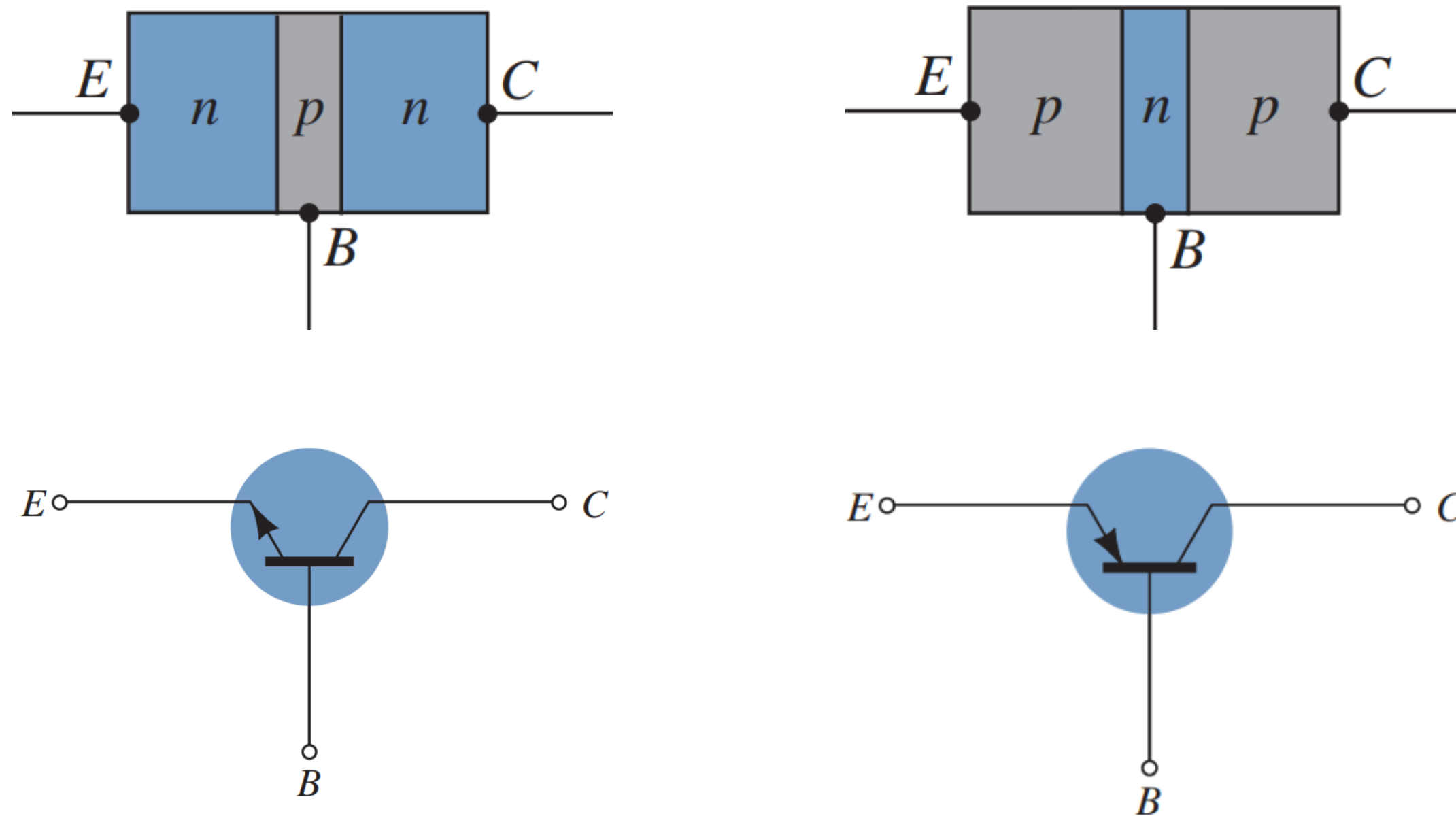
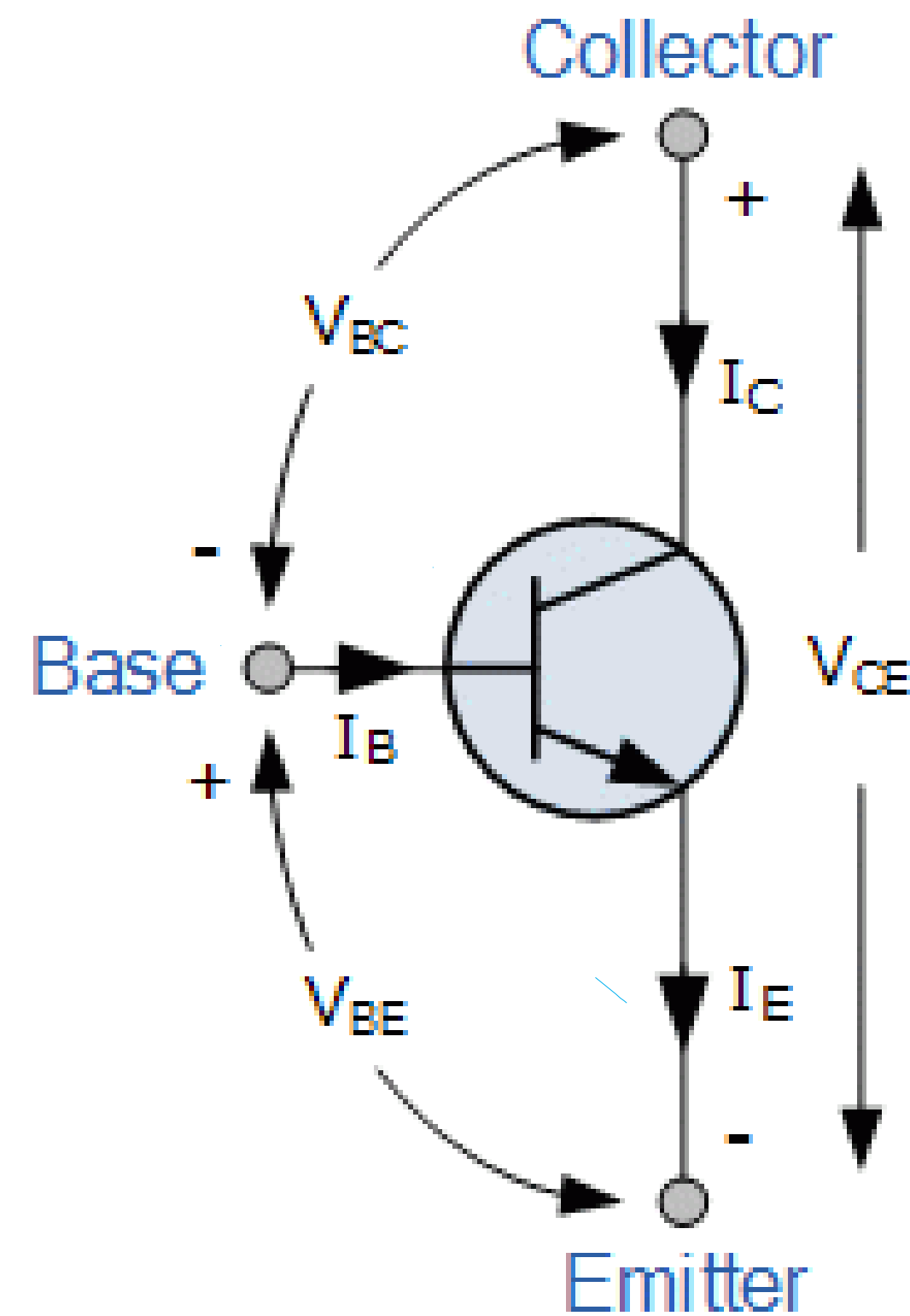


Fig: NPN and PNP Transistor

- A current-controlled semiconductor device
→ A small base current I_B controls a much larger collector current I_C
- Has three terminals – Emitter (E), Base (B) and Collector (C)
- Comes in two types – NPN and PNP
- Two main application – Amplification and Switching

**Why is it called “bipolar”?

BJT: Modes of Operation



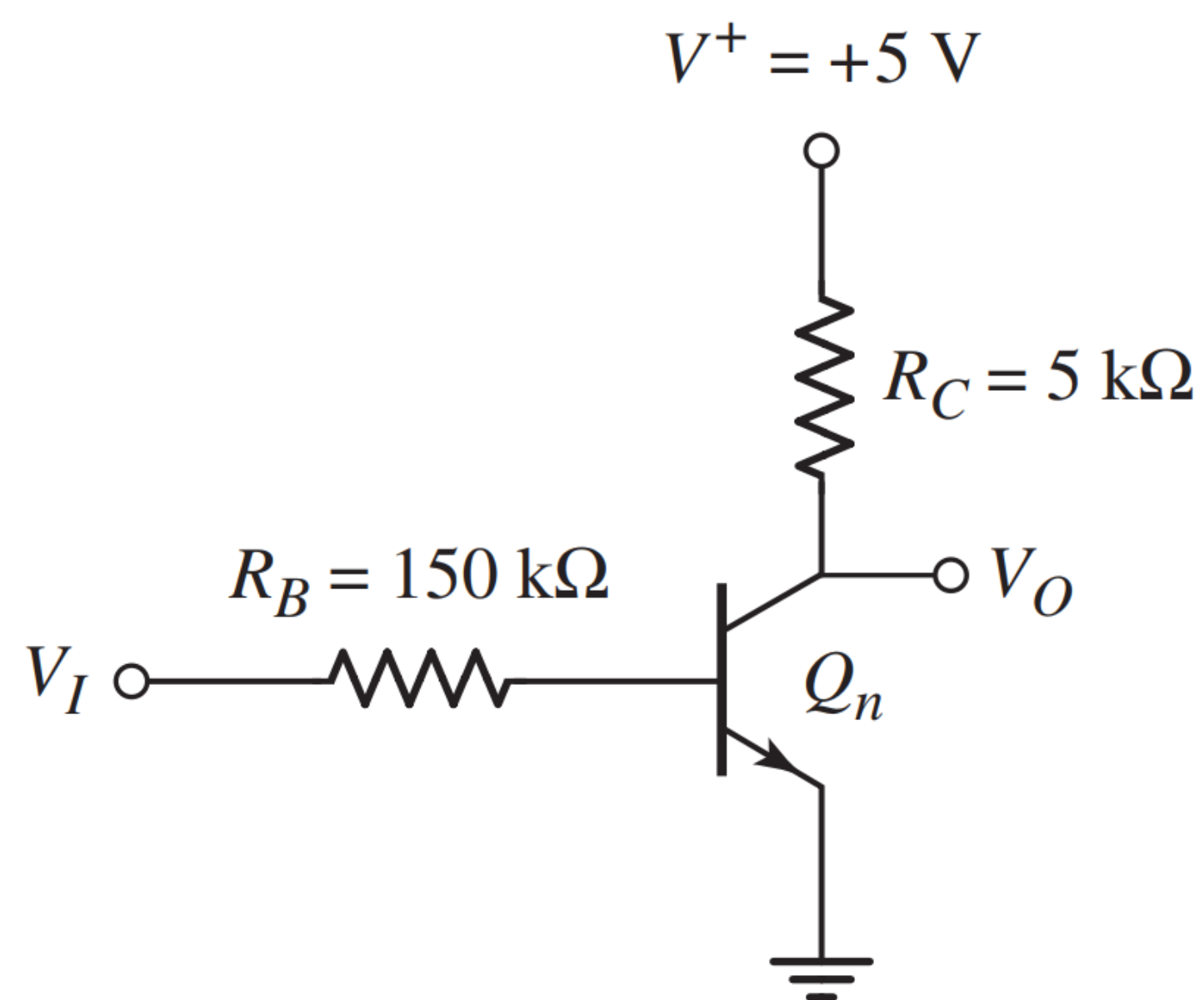
Modes	EB Junction	CB Junction	Application
Cutoff	Reverse	Reverse	Switching
Saturation	Forward	Forward	
Active	Forward	Reverse	Amplifier
Inverse-active	Reverse	Forward	??

Current Relationships	Voltage Relationships	
	Active Mode	Saturation Mode
$I_E = I_B + I_C$ $I_C = \beta I_B$ $\beta = \text{current gain}$ (typically 20-200)	$V_{BE} \approx 0.7 \text{ V}$ $V_{CE} > V_{BE}$	$V_{BE} \approx 0.7 \text{ V}$ $V_{CE} \approx 0.2 \text{ V}$

Voltage Transfer Characteristics of BJT



Objective: Develop the voltage transfer curves for the circuits shown in the figure below. Assume npn transistor parameters of $V_{BE}(\text{on}) = 0.7 \text{ V}$, $\beta = 120$, $V_{CE}(\text{on}) = 0.2 \text{ V}$ (Neamen, Page 311)



Solution:

■ For $V_I \leq 0.7 \text{ V}$:

Transistor Q_n is cut off, so that $I_B = I_C = 0$. The output voltage is then $V_O = V^+ = 5 \text{ V}$.

■ For $V_I > 0.7 \text{ V}$:

Transistor Q_n turns on and is initially biased in the active mode. We have –

$$I_B = \frac{V_I - 0.7}{R_B}$$

- For $V_I > 0.7 \text{ V}$:

Transistor Q_n turns on and is initially biased in the active mode. We have –

$$I_B = \frac{V_I - 0.7}{R_B}$$

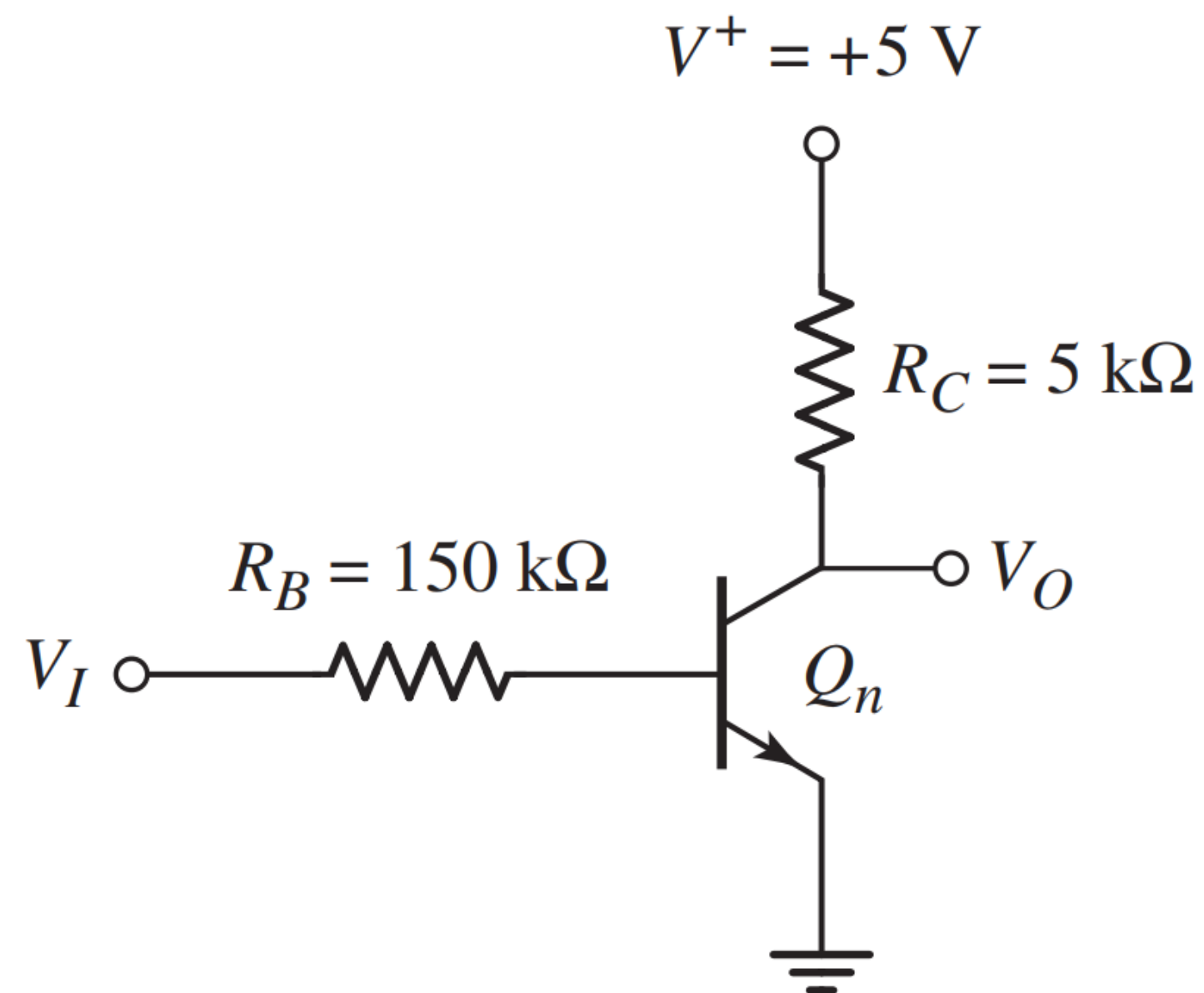
And,

$$I_C = \beta I_B = \frac{\beta(V_I - 0.7)}{R_B}$$

Then,

$$V_O = 5 - I_C R_C = 5 - \frac{\beta(V_I - 0.7)R_C}{R_B}$$

This equation is valid for $0.2 \leq V_O \leq 5 \text{ V}$. When $V_O = 0.2 \text{ V}$, the transistor Q_n goes into saturation.



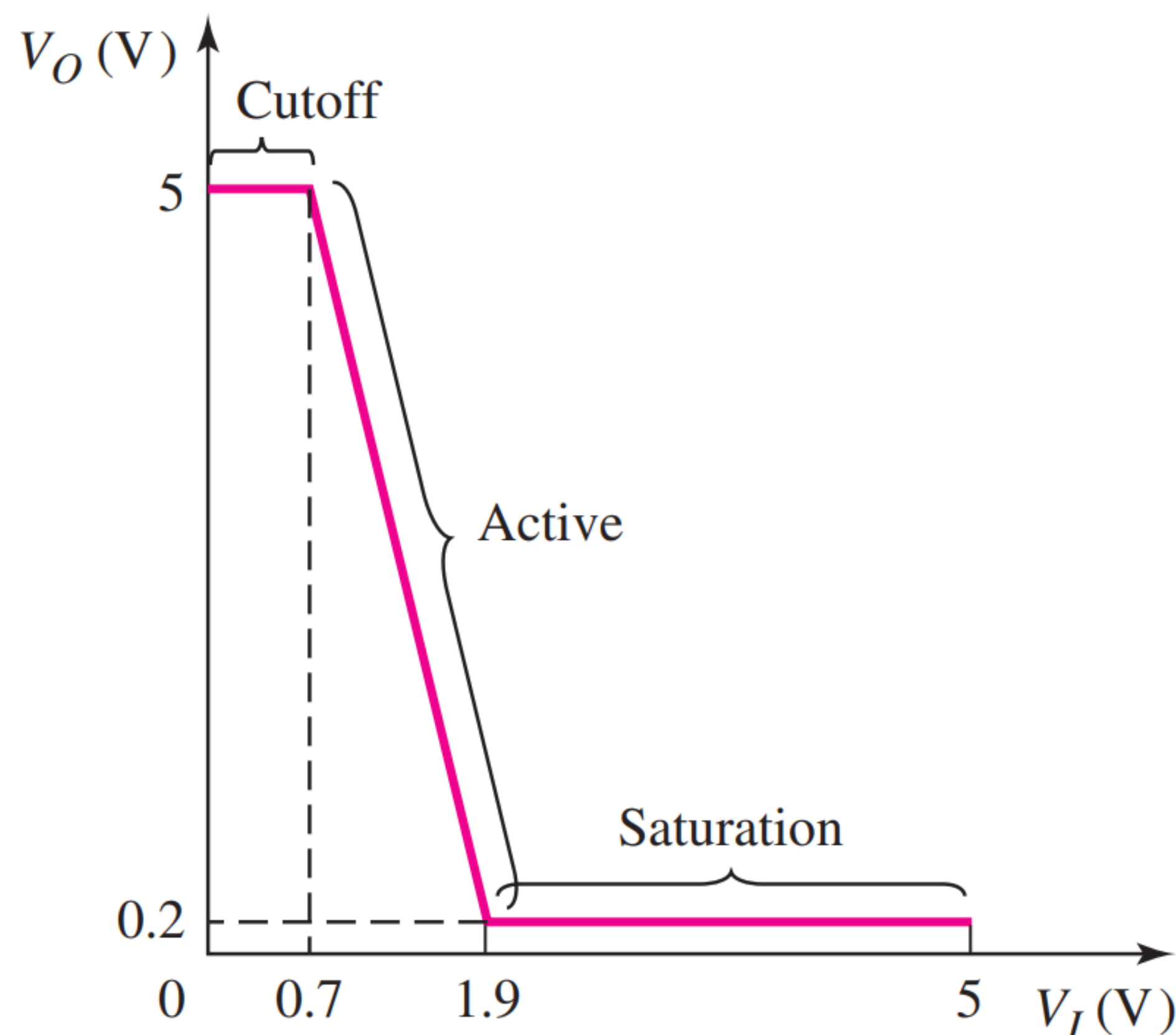


Fig: Voltage transfer characteristics

- For $V_O = 0.2$ V:

When $V_O = 0.2$ V, transistor Q_n goes into saturation. When $V_O = 0.2$ V, the input voltage is found from –

$$0.2 = 5 - \frac{(120)(V_I - 0.7)(5)}{150}$$

Which yields $V_I = 1.9$ V. For $V_I \geq 1.9$ V, the transistor Q_n remains biased in the saturation region.

The voltage transfer curve is shown in the left Figure.

Field Effect Transistor (FET)

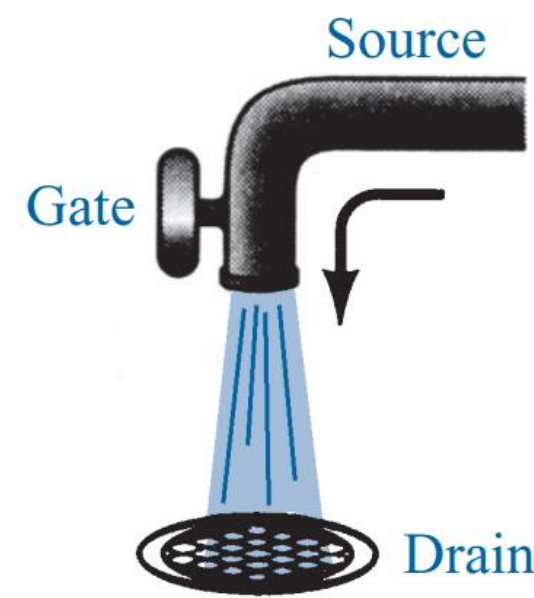
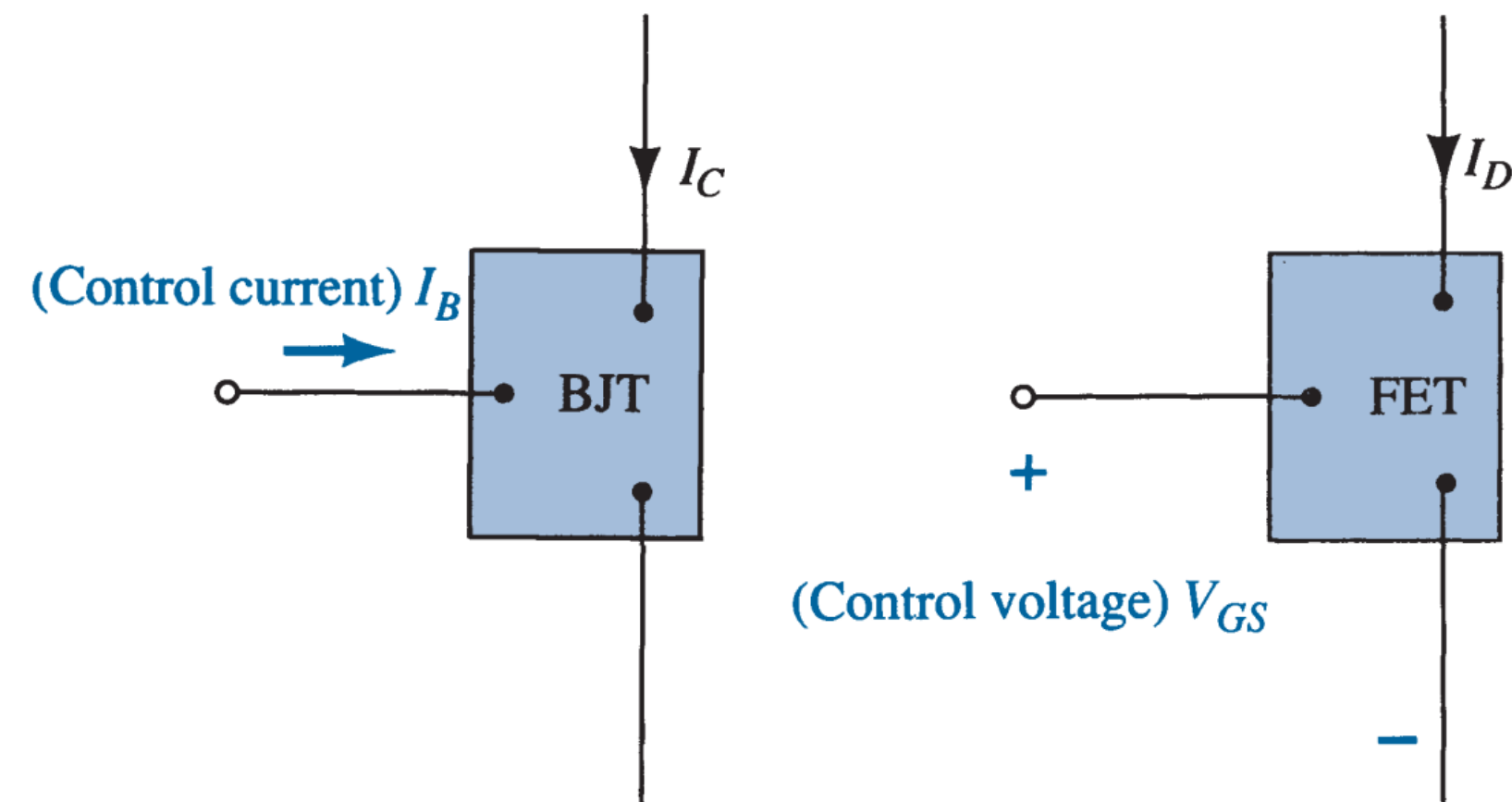


Fig: Field Effect Transistor

Field Effect Transistor (FET) is a three-terminal, **voltage-controlled** device. FET is comparable to BJTs but differs in control mechanism and carrier type.

Key Differences between **FET** and **BJT** –

Feature	BJT	FET
Control Type	Current-controlled	Voltage-controlled
Conduction	Bipolar	Unipolar
Input Impedance	Low	Very high ($M\Omega$ range)
Voltage Gain	High	Lower than BJT
Temp. Stability	Less stable	More stable

There are two main types of FETs: **JFET** and **MOSFET**.

In this course, we will focus on **MOSFET**, as it is widely used in **digital logic gates** (specifically enhancement-type MOSFETs).

MOSFET → **Metal-Oxide-Semiconductor Field-Effect Transistor**

- **Metal-Oxide** Refers to the gate structure, which consists of a metal layer separated from the semiconductor by an insulating oxide layer, typically silicon dioxide.
- **Field-Effect** describes how the device controls current using an electric field created by the gate voltage, rather than current injection.

Enhancement-type MOSFET

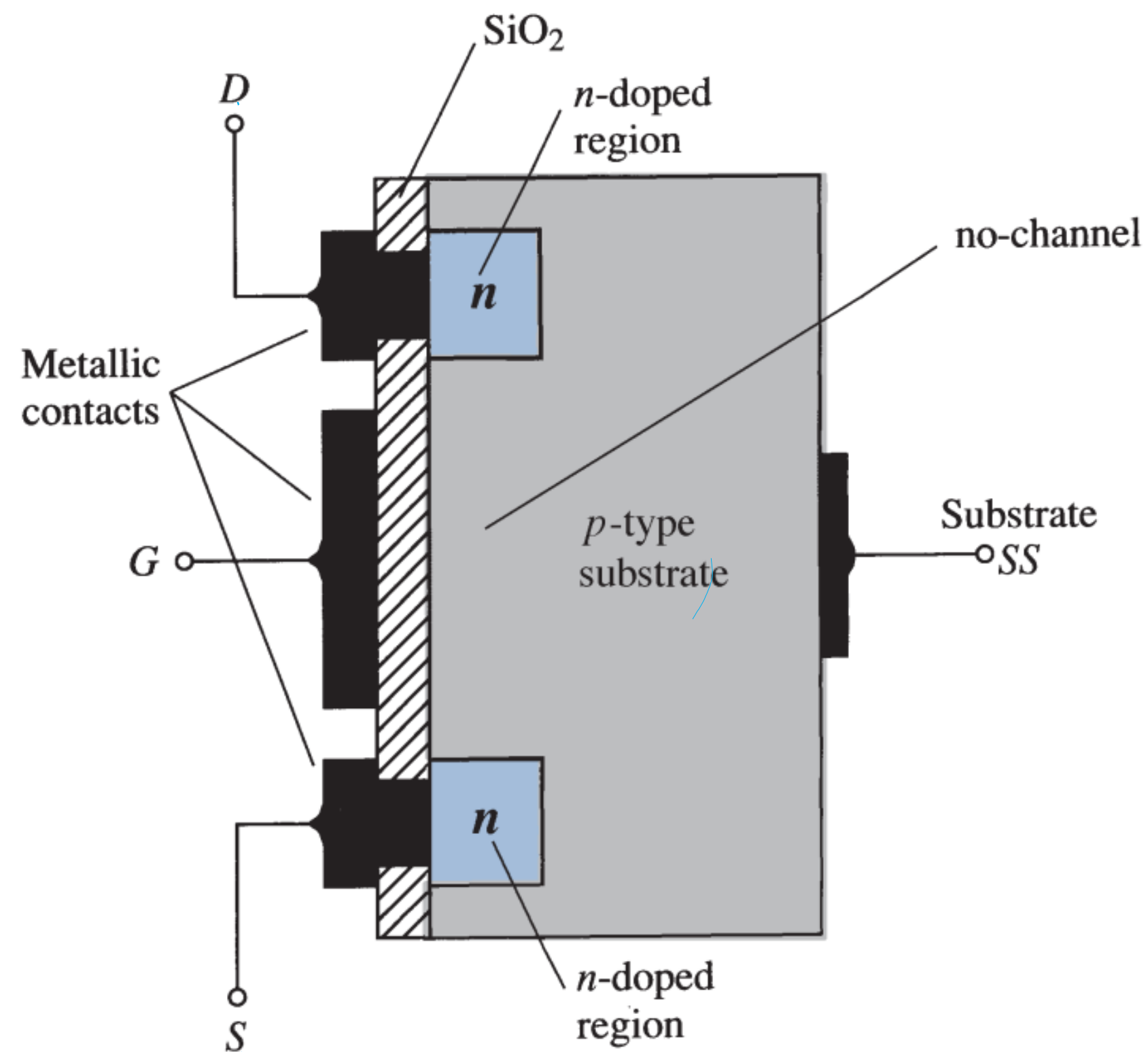


Fig: n-channel enhancement-type MOSFET

- In the left figure: **n-channel enhancement-type MOSFET**
- Built on a **p-type** silicon **substrate**
- **Source (S)** and **Drain (D)** formed by **n-type** material
- **Gate (G)** is insulated from the substrate by a thin **SiO₂** layer
- No physical conducting channel between source and drain at zero gate voltage
- Control of conduction is achieved electrically via gate voltage

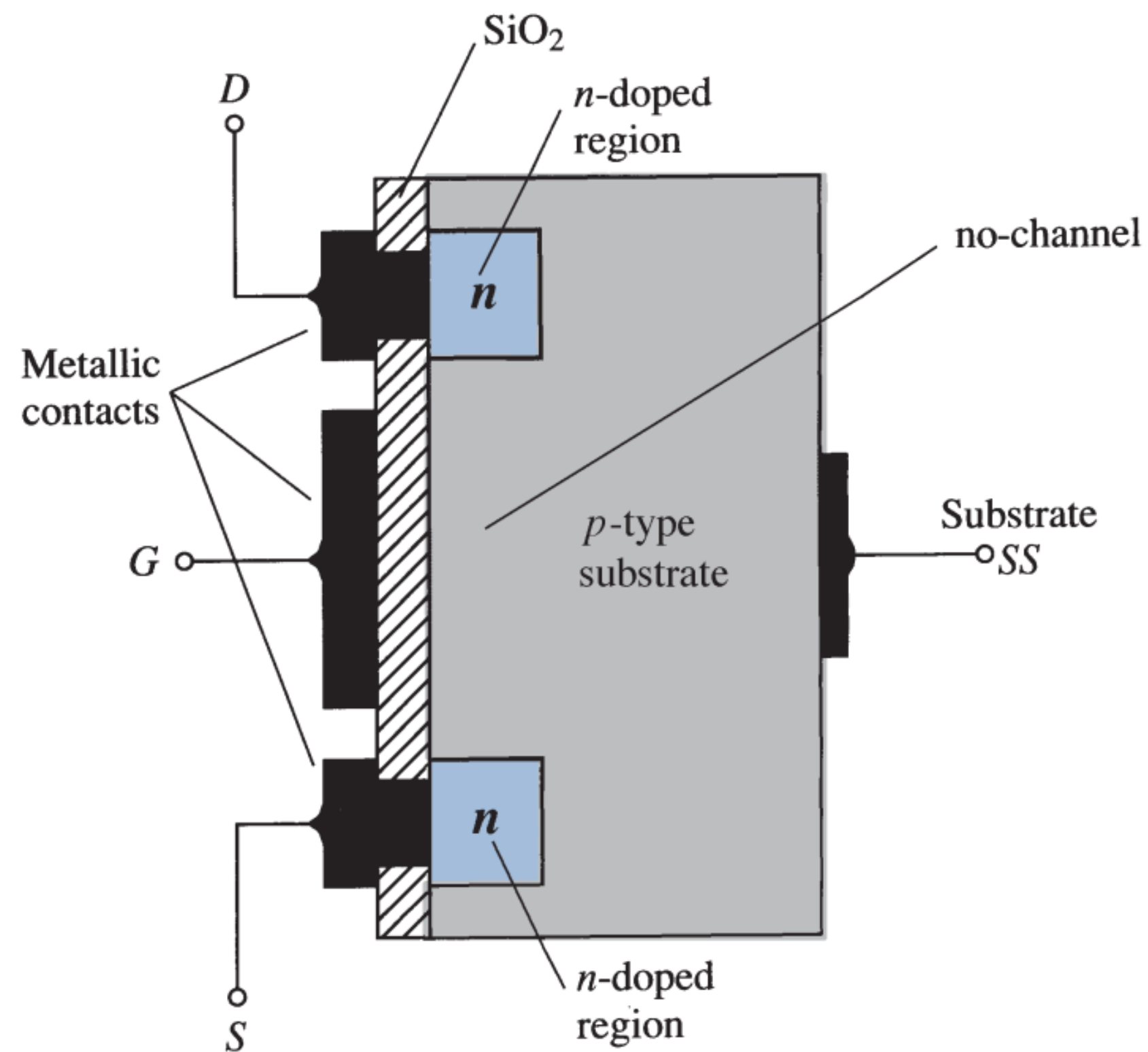


Fig: n-channel enhancement-type MOSFET

Key Construction Features

- **Substrate:**
 - › May be internally connected to Source, or
 - › Externally connected via a fourth terminal (often labeled **SS** or Body)
- In contrast to **depletion-mode MOSFET**:
 - › No channel is pre-built
 - › A channel must be induced by applying a suitable V_{GS}
- **SiO₂** insulates Gate from underlying semiconductor
→ extremely **high input impedance**

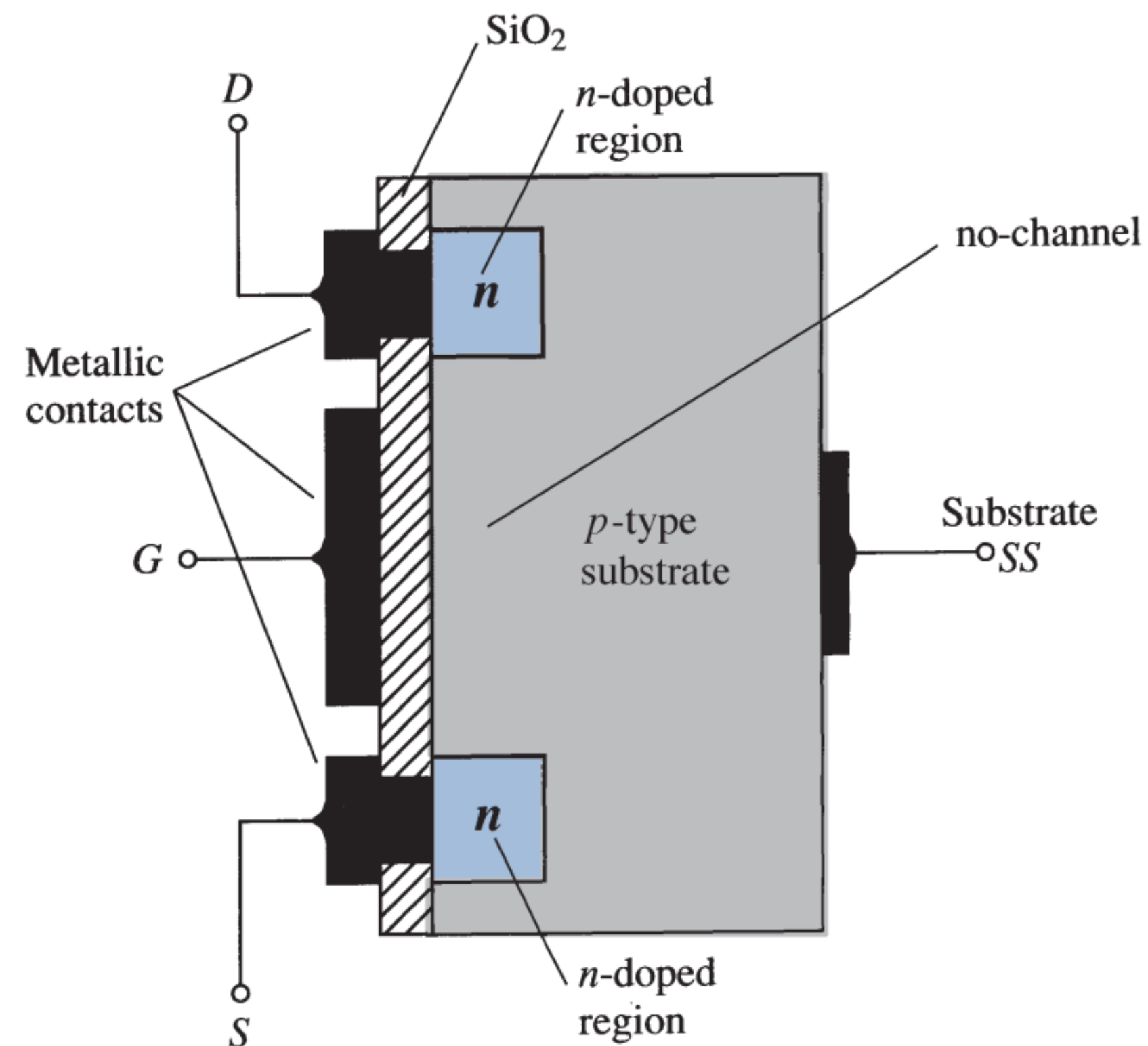


Fig: n-channel enhancement-type MOSFET

Behavior at $V_{GS} = 0\text{ V}$

- Apply a positive V_{DS} , but with $V_{GS} = 0\text{ V}$:
 - > No channel $\rightarrow I_D \approx 0\text{ A}$
- Electrons exist in n-doped Source and Drain region, but:
 - > No conduction path due to **absence of channel**
 - > Two reverse-biased p-n junctions block current flow

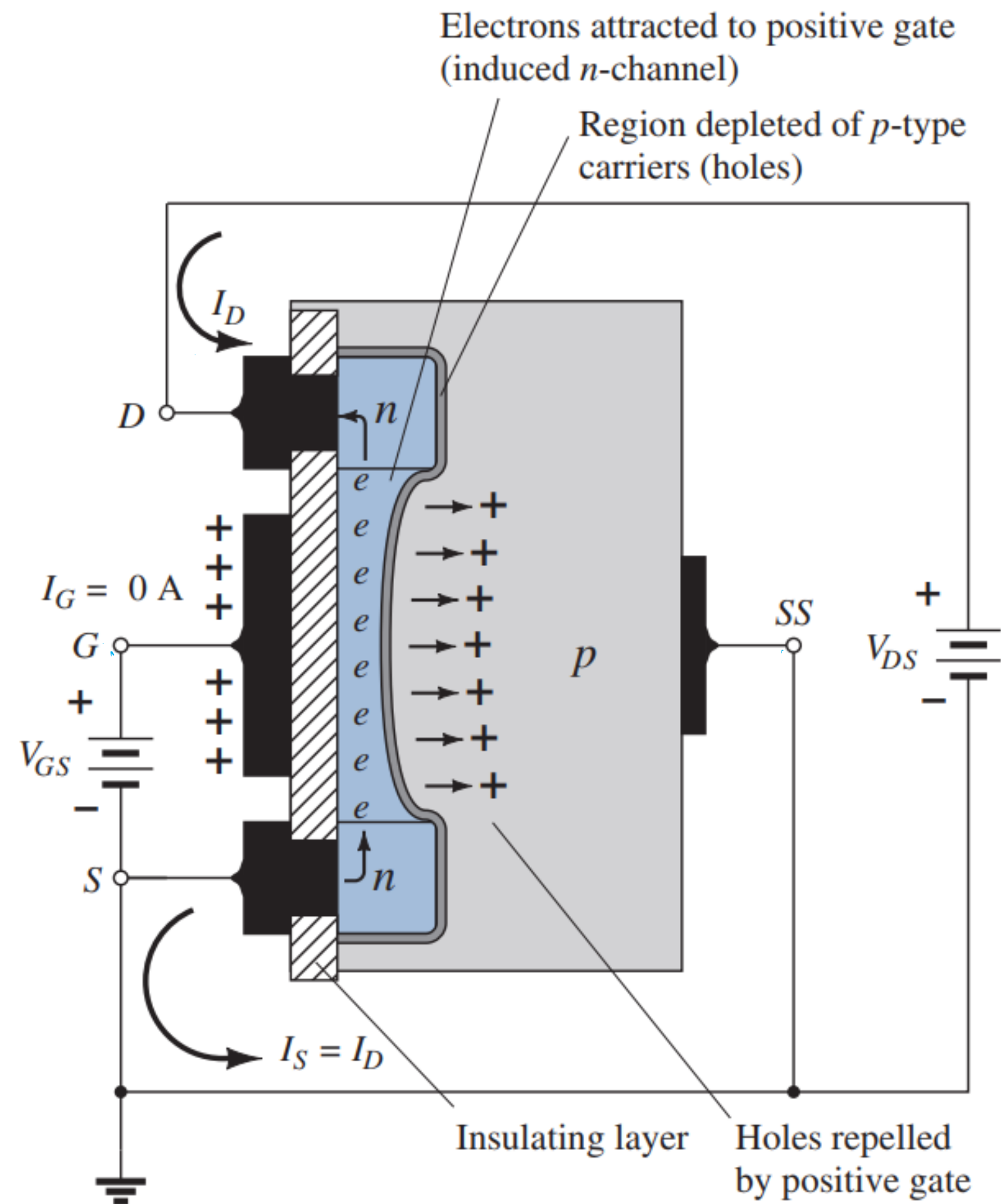


Fig: n-channel enhancement-type MOSFET

Behavior at $V_{GS} > 0$ V

■ When $V_{GS} > 0$:

- > Positive gate voltage repels holes (majority carriers) along the edge of the SiO_2 layer to leave the area and enter deeper regions of the p-substrate
- > Electrons are attracted and (minority carriers) accumulated near gate
- > An inversion layer (n-channel) forms, enabling conduction
- > **Threshold Voltage (V_T or $V_{GS(th)}$):** Minimum gate voltage to form a channel

Enhancement-type MOSFET

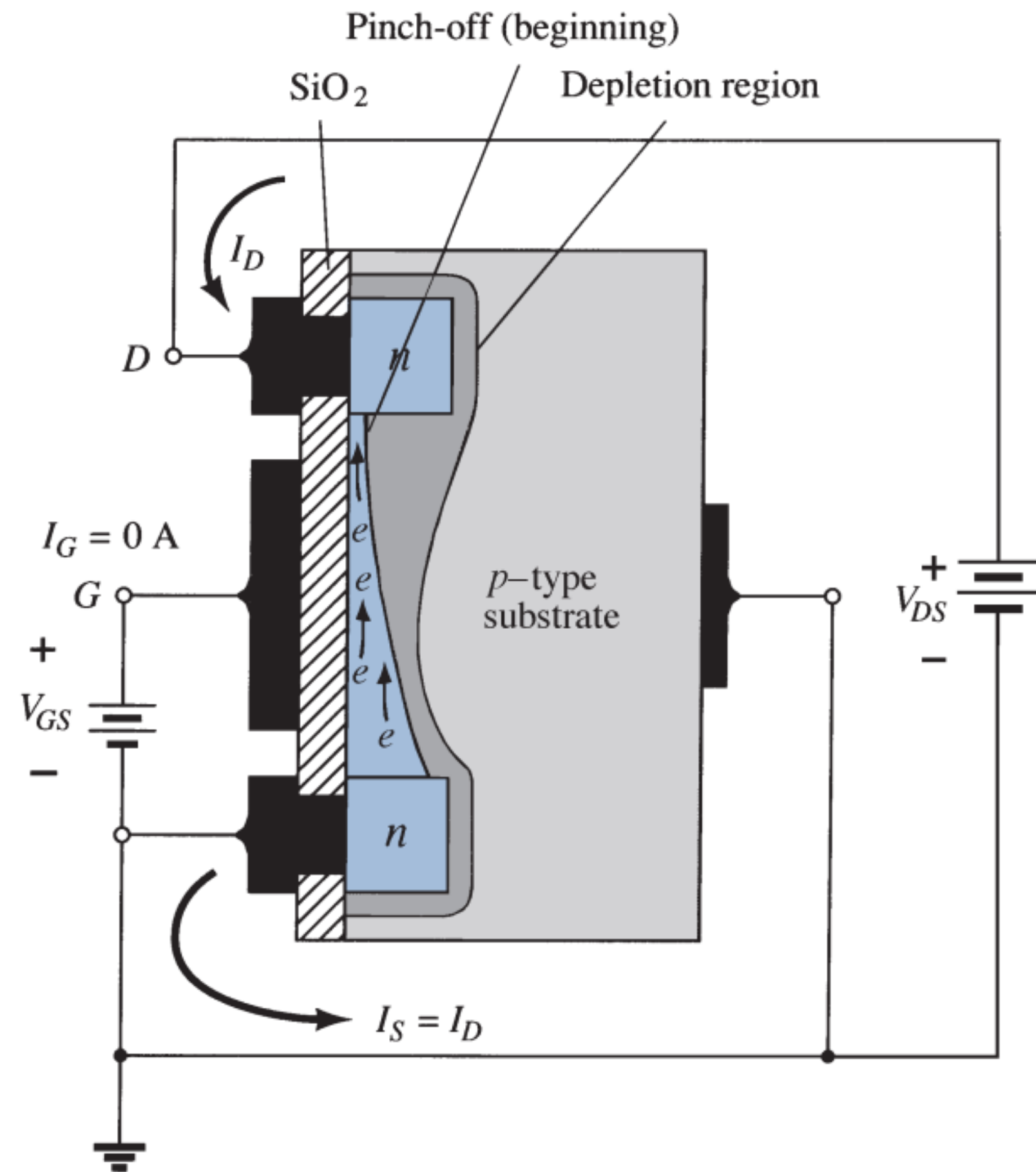


Fig: n-channel enhancement-type MOSFET

Behavior at $V_{GS} > 0$ V

■ Enhancement Mode Operation:

- > Channel forms only after V_{GS} exceeds V_T
- > Thus, called “Enhancement-type”
- > ID current increases as V_{GS} increases beyond V_T

■ Saturation and Pinch-Off:

- > If V_{GS} is fixed, increasing V_{DS} causes V_{GD} to decrease, causing channel near Drain to **pinch off**
- > Beyond this, I_D saturates despite increasing V_{DS}

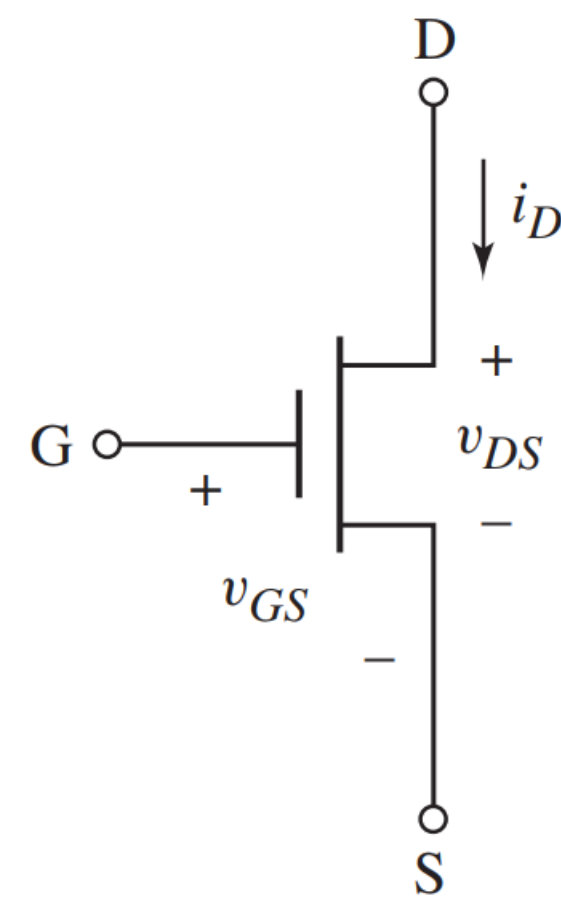
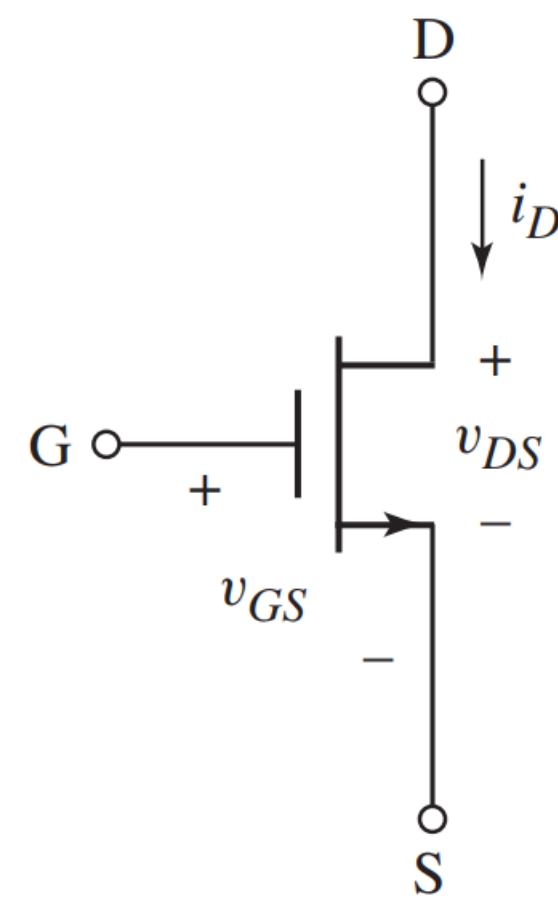
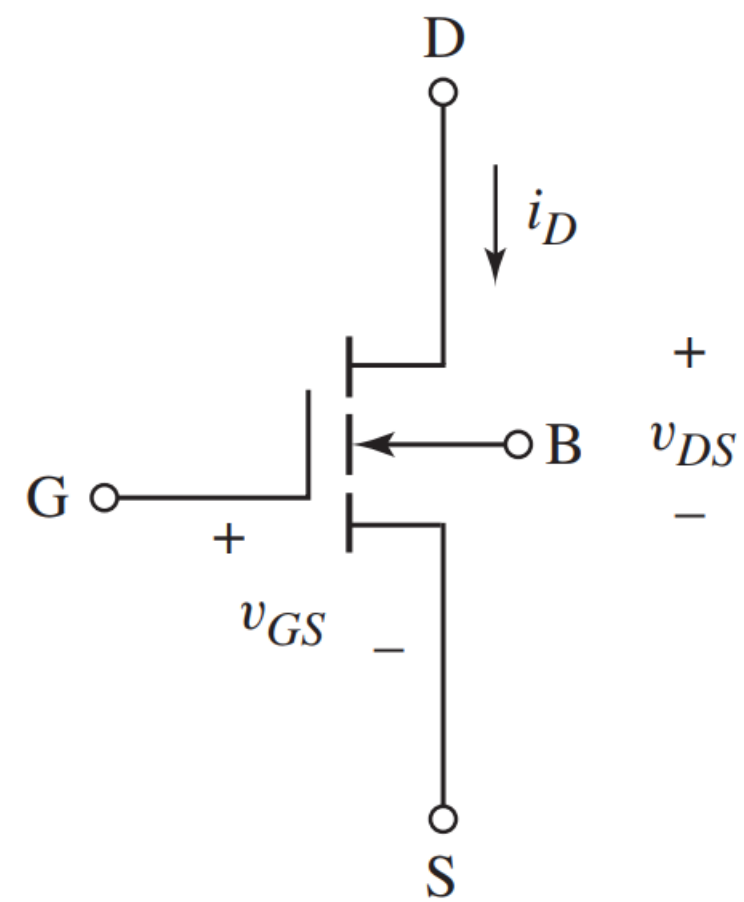
Before pinch-off, increasing V_{DS} does two opposite things:

1. Stronger drain-source field $\rightarrow$ pulls electrons harder $\rightarrow$ tends to increase current.
2. Channel thinner near drain $\rightarrow$ more resistance $\rightarrow$ tends to reduce current.

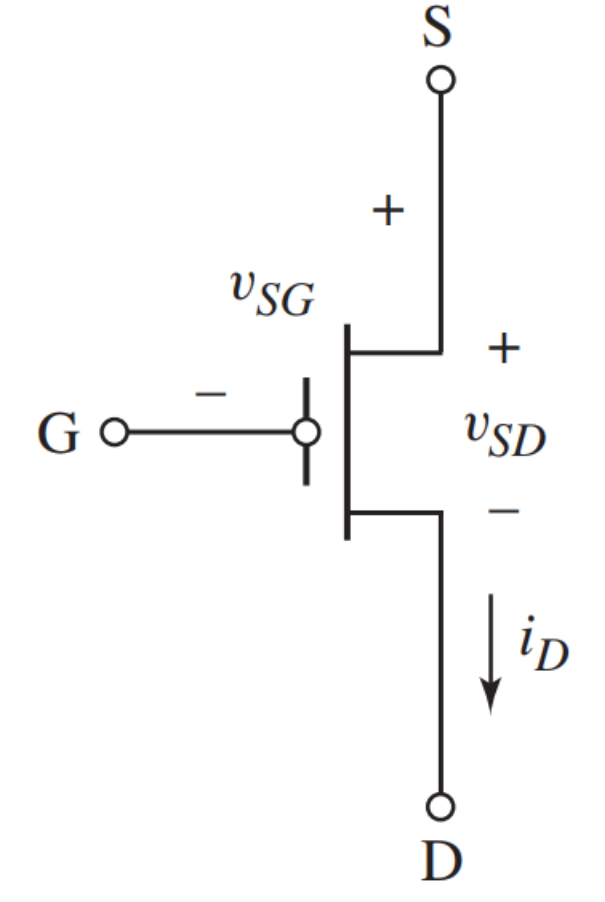
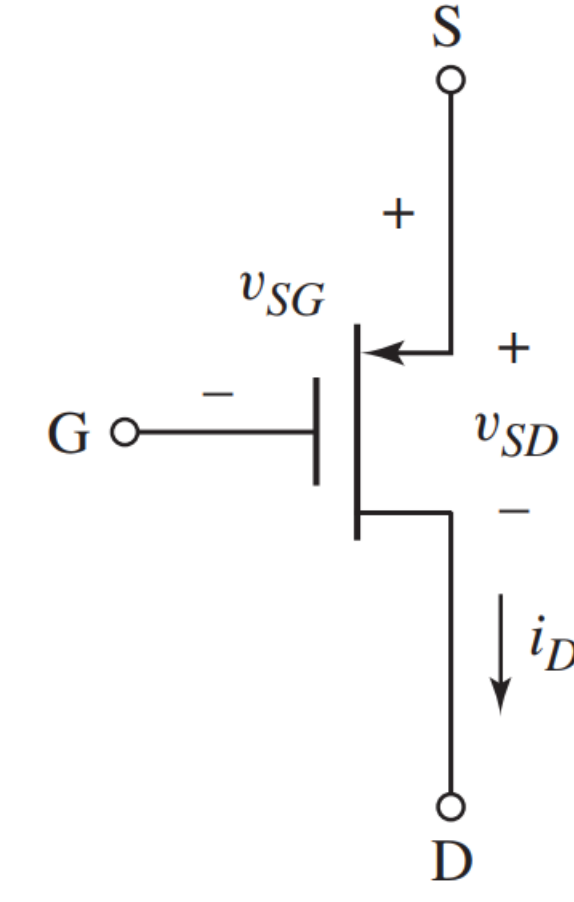
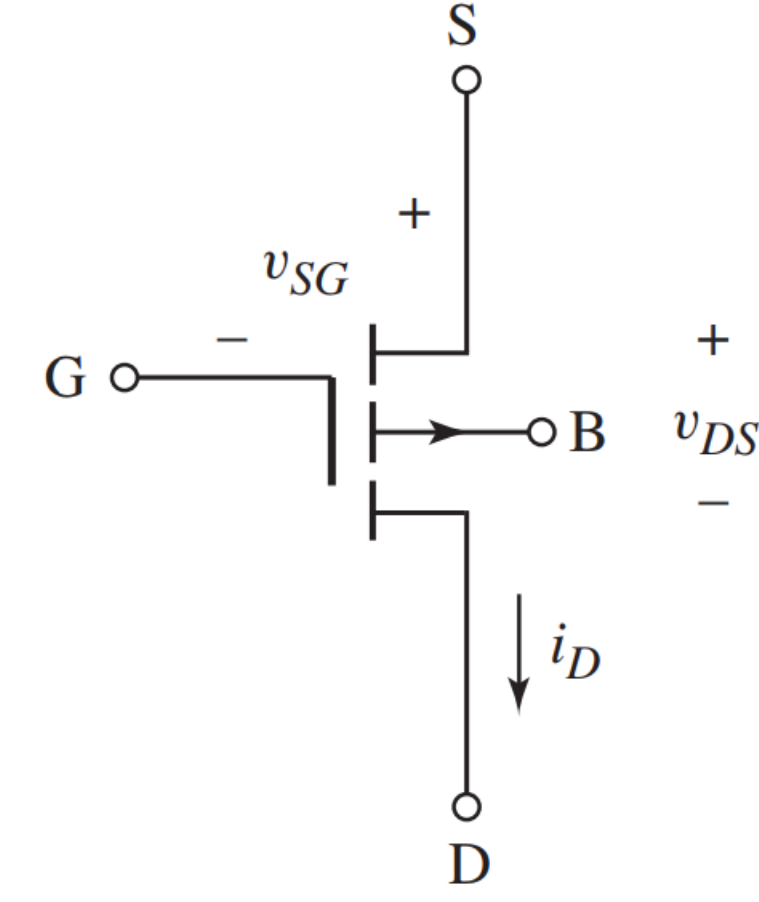
Before saturation, effect 1 dominates, so I_D increases.

At pinch-off, the drain end of the channel becomes extremely thin, but the current does not stop because electrons that reach the pinch-off point are swept quickly across the depletion region into the drain by the strong electric field. After pinch-off, increasing V_{DS} mostly increases voltage across the pinch-off/depletion region, not across the main conducting channel. So the “useful” part of the channel sees almost the same voltage: $V_{DS(sat)} = V_{GS} - V_T$

Commonly used Circuit Symbols



n-channel enhancement-type MOSFET



p-channel enhancement-type MOSFET

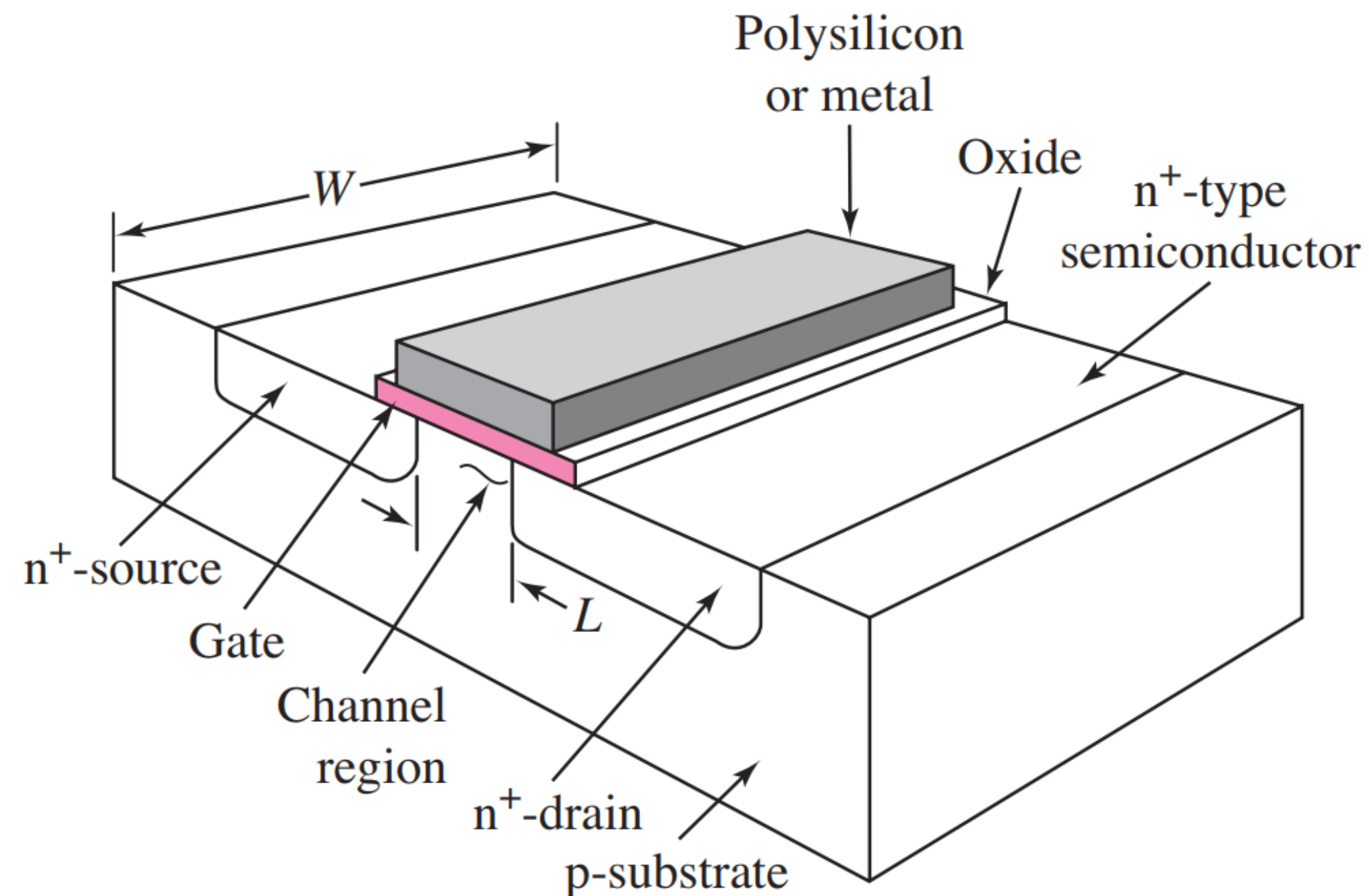


Fig: Structural diagram of an n-channel enhancement-type MOSFET

- Enhancement-type MOSFETs are the building blocks of digital logic –
 - › n-channel enhancement MOSFET → **NMOS**
 - › p-channel enhancement MOSFET → **PMOS**
- **CMOS (Complementary MOS):**
 - › Combines NMOS + PMOS
 - › Offers low power, high noise margin, high density
 - › Used in nearly all modern digital logic ICs